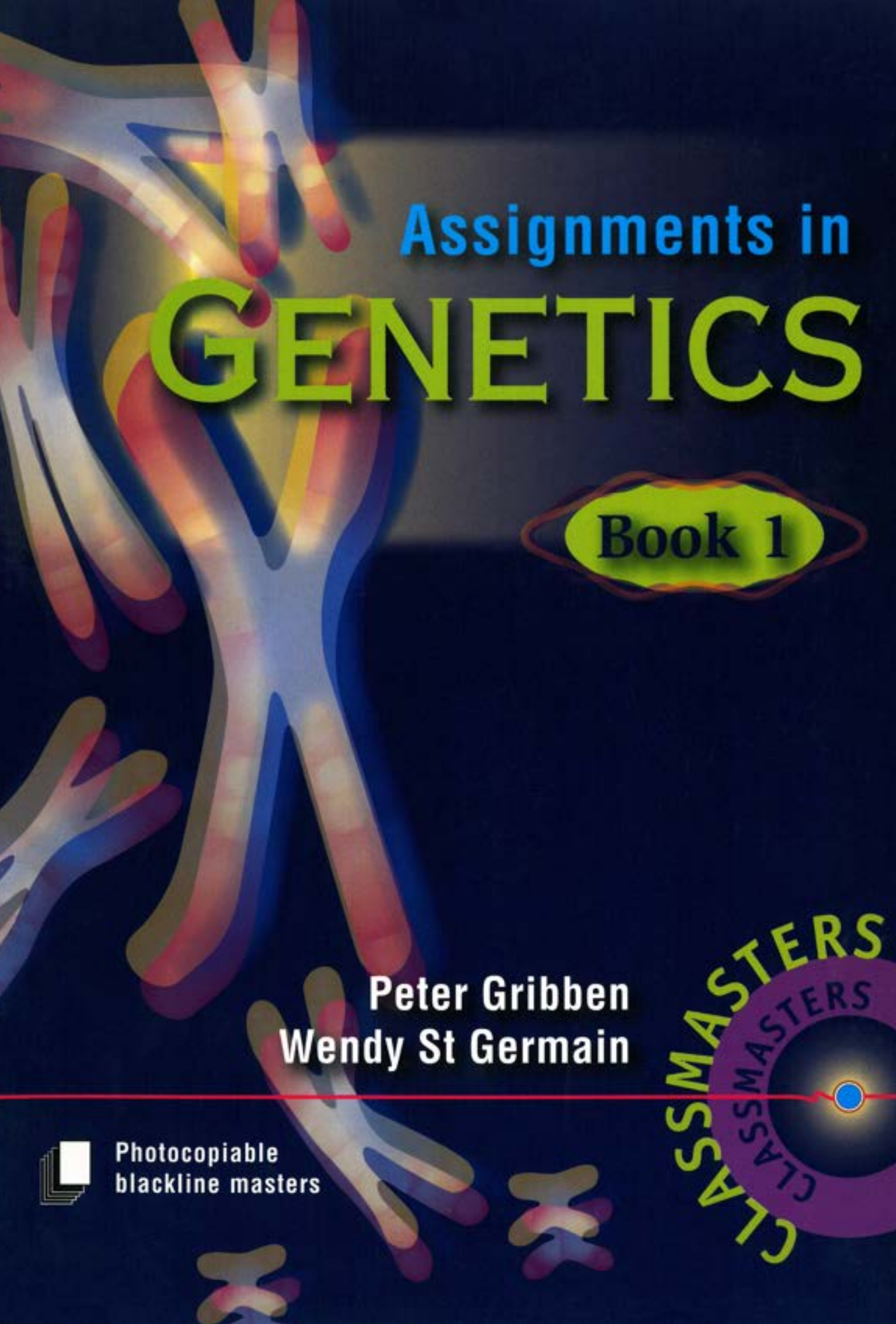




Assignments in **GENETICS**

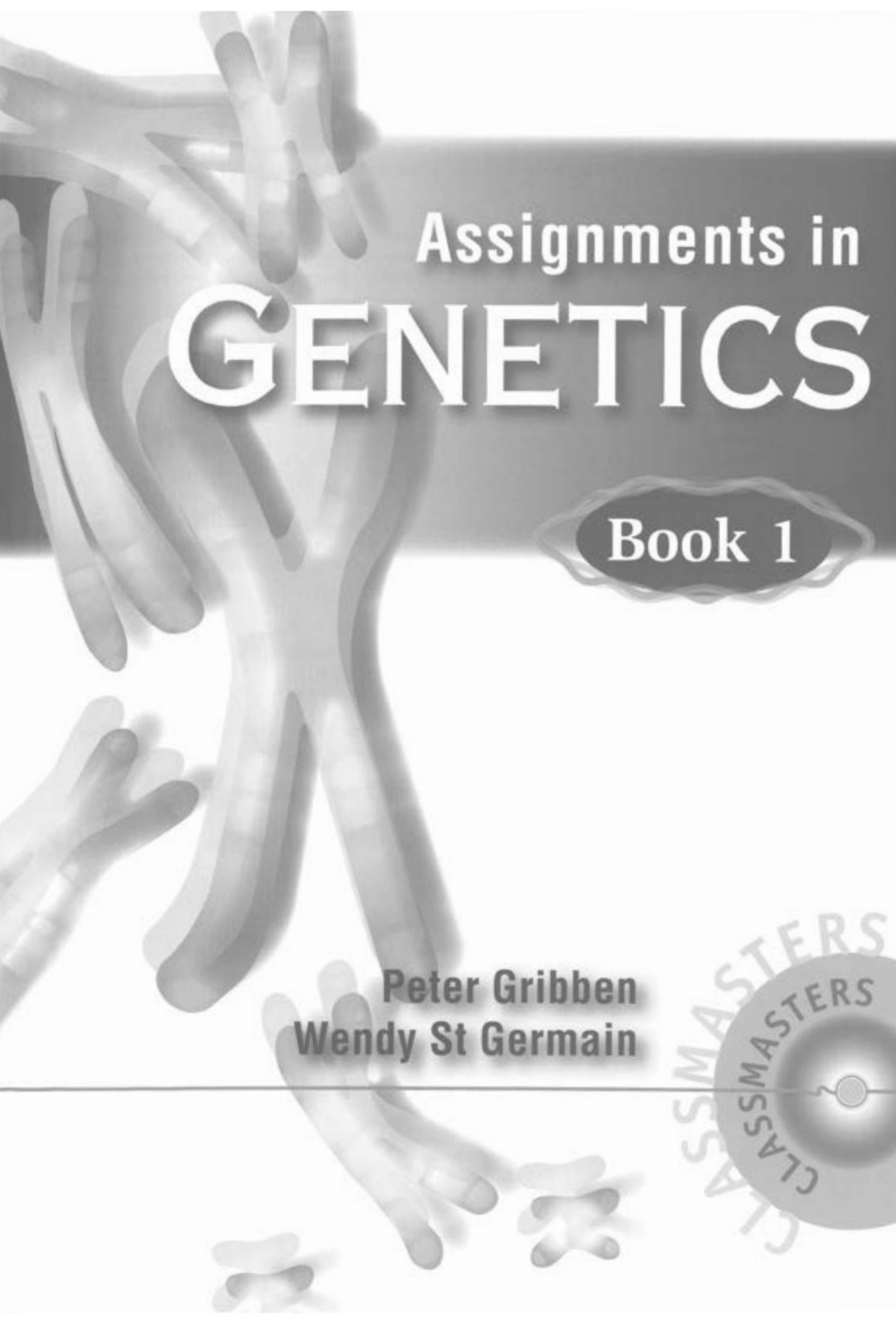
Book 1

**Peter Gribben
Wendy St Germain**



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Assignments in **GENETICS**

Book 1

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Introduction

Genetics is a topic that is increasingly of interest to teachers, students and the general public. It has become an important part of high school science syllabuses, both in junior and senior courses. *Instant Lessons in Genetics book 1* has been written with the needs of middle high school students in mind, however, teachers of senior classes will be able to use the material for revision and consolidation with their classes.

Instant Lessons in Genetics book 1 has been specifically written so that the extensive technical terms and concepts used in Genetics are introduced through a variety of activities, making it easier for students to understand and remember each concept. In keeping with this, a considerable portion of this book contains 'stand alone' exercises, requiring little or no previous knowledge of the subject. Other sections of this book can be used to assess each student's progress and knowledge of Genetics, and to identify areas of weakness. Selected answers have been provided for each work sheet.

Instant Lessons in Genetics book 1 builds student knowledge, confidence and interest in a fascinating, if occasionally challenging subject.



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Evolution: a theory describing the processes that have transformed life on Earth from its earliest forms to the great diversity of organisms existing today.

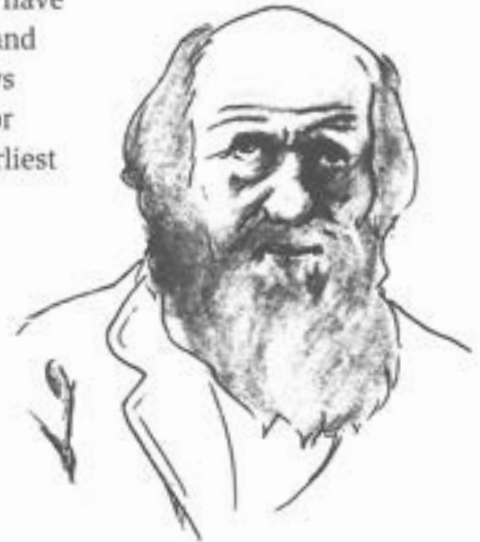
The vast majority of scientists use the *Theory of Evolution* to explain how the species of plants and animals that live (and have lived) on the Earth have come about. By understanding why species have evolved, scientists can predict, with some accuracy, how outside factors might affect future generations. The theory also helps scientists to better understand relationships between fossils and to learn more about our past.

- 1 Explain what is meant by the term 'fossil'.

Naturalists and scientists have long believed that species changed, but were uncertain as to why the changes occurred.

- 2 'A species can be described as a group of similar organisms that can breed with each other.' True or false?

Charles Darwin (1809–82) questioned many things, such as why some insects were rare and others were common. He observed that cows introduced to graze in a field applied a *directional pressure* on grasses growing where they grazed. Although the whole area may have started with an identical mix of, for example, quick growing and slow growing grasses, the ratio of grasses began to change if cows were present. This was because the cows selectively ate (or ignored) certain species of grass. This was one of the earliest documented examples of *selective pressure*.



- 3 Which type of grass would do better in a field grazed by cows, a quick-growing grass or a slow-growing grass? Explain your answer.
- 4 Suggest one other factor that might influence the ratio of grasses present if cows were present.
- 5 In your own words, describe what is meant by 'selective pressure'.
- 6 Give another example of selective pressure.

Darwin was a very observant person and he was the first to suggest that the pressures of nature could affect how species changed over time.

- 7 Write the name of a common plant or animal and give three examples of natural pressures that could affect this species' ability to grow and reproduce.
- 8 Write the name of a common plant or animal and give three examples of directional pressures, applied by humans, that could affect this species' ability to grow and reproduce.

Darwin believed that some individuals in a population possessed characteristics slightly different from those of other members. These differences might better enable some individuals to survive than the rest. Those who were better equipped to compete in their surroundings survived and reproduced in greater numbers. The offspring ('children') of the successful parents inherited the traits (characteristics) that made the previous generation better survivors. Because more young of successful parents were born, the percentage of the population possessing these traits increased. It is possible that Darwin was influenced by his grandfather's book, *Zoonomia*. This book, which wasn't a scientific text but more a book of musings, had a central theme of how the environment affects an organism's survival.

- 9 Give a brief definition of 'environment'.



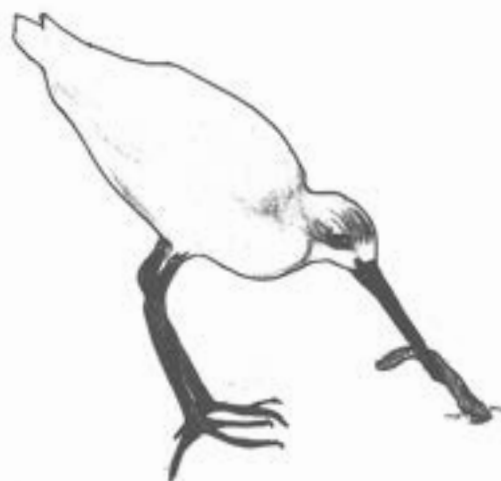
Various pressures can be applied on areas of land.

- 1 Decide which of the actions described below apply directional pressure on the piece of land described? Explain your answers.

- A certain naturally occurring plant species is removed from a paddock whenever it appears.
- Planting annuals (complete life cycle in one year) in a garden every summer.
- Weekly spraying of insecticide in an orchard.

Population: all the members of a particular species (related organisms) found in an area.

Consider a small population of 20 birds (10 male and 10 female). Due to a change in the environment their normal food sources (berries and caterpillars) are growing scarce. However, due to these changes, a new abundant food source is available (mud worms). In order to reach the worms the birds must have long, probing beaks. Long beaks are not common in this population, only four birds (two of each sex) have them.



- 2 Using this population as an example, show how a species may evolve. Use the following assumptions over three generations:
- five short-beaked birds die, or do not reproduce.
 - the surviving birds reproduce
 - successfully reproducing birds produce three young each
 - long beaks are a dominant feature (only one dose of the gene is required for the offspring to express it)
- 3 What fact about the genomes (genetic material) of the four original long-beaked birds is unknown here?
- 4 With regards to your answer to Question 3, how might your evolution predictions be affected?

'Individuals can be selected but populations evolve.'

- 5 Discuss the statement above, with reference to the bird population mentioned previously.

Alfred Wallace

Working at about the same time as Darwin, Alfred Russel Wallace (1823–1913) came up with similar ideas about evolution. Wallace explained selective pressures using animal survival as his example. Those animals who can hunt more successfully, and cope better with difficulties in their environment, will survive illness and shortage of food, when others fail. By doing so they will live to reproduce, giving to their offspring the traits that made them better able to cope. This idea is now commonly known as 'survival of the fittest'.

- 6 Give a brief definition of 'trait'.
- 7 Give a brief definition of 'offspring'.
- 8 Briefly describe, by reference to a particular animal, an example of *survival of the fittest*.



The Theory of Evolution

3

Alfred Wallace was an English naturalist who independently arrived at similar conclusions to Charles Darwin concerning the theory of natural selection, often referred to as 'survival of the fittest'.

- 1 Study the definition of natural selection given below and rewrite it in your own words.

Natural selection: A process resulting in the survival and reproductive success of organisms best adjusted to a specific environment, and that leads to the passing on of genetic qualities best suited to that environment.

Darwin had worked out most of his theories by the 1840s but was reluctant to publish them, perhaps aware of the controversy they would cause. Wallace wrote to Darwin in 1858 setting out his theory of natural selection as a mechanism for evolution, which was virtually identical to Darwin's ideas. Although Darwin suggested that the two present their work together, Wallace graciously declined. He allowed Darwin to publish his work first, on the grounds that Darwin had spent many years collecting evidence with which to support it. One viewpoint is that Darwin cheated Wallace of recognition for his work, but today we acknowledge the efforts of both men. Wallace was also spared much of the religious backlash Darwin experienced.

- 2 Why might some people take exception to Darwin's theories on evolution?
(Hint: look at the quote below by Benjamin Disraeli, the British Prime Minister, in 1864.)

'Is man an ape or an angel? I, my lord,
I am on the side of the angels.'



Copy the following passage on the *Theory of Evolution* and fill in the missing words:

Parents produce more (3) than can survive. Not all offspring are (4) to their parents. The variation between individuals will make some better able to (5) the pressures of nature than others. The (6) (features) that make the individual better suited to survive will be passed on to their offspring. Because many will possess this trait, over time it will (7) in the population. Those without it will (8) or produce fewer surviving offspring.

Food chains and food pyramids

Food chains and food pyramids, respectively, show the links between 'who eats who' and the relative numbers involved.

- 9 Which type of species in a food chain will produce large numbers of offspring and which will produce fewer?
10 Why do you think evolution has produced the result you gave as your answer for Question 9?

Darwin was only able to explain evolution on a theoretical basis. The true field of genetics was not born until the work of Gregor Mendel, an Austrian monk and contemporary of Darwin, was made public.

- 11 Explain what is meant by 'theoretical'.
12 Explain what is meant by 'contemporary'.



Genetics: the study of heredity, or the 'passing on', of characteristics from parents to offspring.

The work of the Austrian monk Gregor Mendel (1822–84) has led to the understanding that traits are passed down from parent to offspring in little packages called *genes*. Mendel was a botanist, a monk and a teacher. A monastery garden and a variety of peas served as the laboratory and materials for his work on the first laws of genetics.

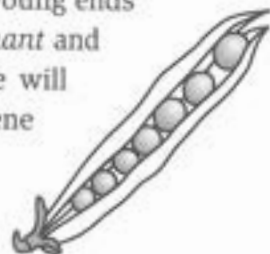
1 What is a 'botanist'?

Mendel observed that his pea plants produced either yellow seeds or green seeds. When he crossed plants producing yellow seeds with those producing green seeds, he discovered that the offspring all produced yellow seeds, not a blend as he expected. He allowed the plants to self-fertilise and, to his surprise, the next generation of peas produced seeds in a ratio of three yellow to one green.

2 What is meant when plants are described as 'self-fertilising'?

After many further trials, the same ratio of 3 : 1 appeared. Mendel decided that the colour was determined by a tiny unit of information now recognised as an *allele*. Mendel reasoned that some of the new generation peas had inherited green 'information' from yellow parents. This showed that the alleles must be in pairs, together forming one hereditary unit (now called a *gene*). The young ends up with two sets, one set from each parent. From this came the concept of *dominant* and *recessive* genes. If two non-identical genes are present, then the dominant one will prevail over the recessive one and only the characteristic of the dominant gene would appear in the offspring.

3 Give two reasons why Mendel might have chosen peas to study.



Mendel's First Law: The Law of Equal Division

Whenever we have one pair of alleles, we must assume that they will separate into two equal halves. Although each parent possesses two alleles for a given trait, when gametes (sex cells) are produced, only one allele from each parent will be given to the offspring.

Mendel's Second Law: The Law of Independent Assortment

When gene pairs separate, each offspring generally has a 50/50 chance of getting one half or the other. In the same way that shuffling and dealing cards gives each player an equal chance of getting any card in the deck, so does independent assortment suggest that the offspring will receive any of the genes available in the genetic 'deck'. By systematically studying one trait at a time, Mendel laid the foundation for modern genetics. In general, the mathematical formulae he developed to support his theories still work and, fundamentally, his laws still apply.



Between 1856 and 1863 Mendel cultivated approximately 30 000 plants, and he published his results in 1866. Those results remained ignored until 1900 and only came to public attention in the late 1920s.

Alleles: variations of a gene for a particular characteristic.

4 What were the alleles for seed colour in the peas studied by Mendel?

5 Which of the colours for seeds was dominant and which was recessive? Give a reason for your answer.

6 Give one reason why Mendel cultivated so many plants.



Genetics is a branch of biology. It is the study of inheritance — the passing on of characteristics from one generation to the next. The term 'genetics' was first used by William Bateson (a British biologist) in 1906.

Although theories of inheritance, or heredity, were put forward at least as early as the 5th and 4th centuries BC, the study of genetics only truly began in the 18th and 19th centuries. Observations were made of how specific characteristics of plants and animals were passed on from one generation to the next, to provide a rational basis for the improvement of crop plants and livestock.



The most significant breakthrough in genetics was made by the Austrian Monk Gregor Mendel (1822–84). He observed specific features of the pea plant and counted the number of individuals in which each characteristic appeared through several generations. By concentrating on just a few features and determining what proportion of each generation received them, he was able to demonstrate specific patterns of inheritance. These patterns have been shown to be applicable to most genetic systems.

There are two ways in which living things reproduce themselves — sexually and asexually. This sheet deals with sexual reproduction, which leads to genetic variation.

All animals and plants which reproduce sexually start with a female sex cell (egg or ovum) which is entered and fertilised by a male sex cell (in animals this is sperm; in flowering plants it is pollen). Sex cells are called *gametes*.

The fertilised egg (*zygote*) grows into a new individual by a series of cell divisions. The new individual is of the same species and has the *characteristics* of its parent cells. The 'recipe' for this is carried in the parent gametes. The structures which cause this *inheritance* of these characteristics are called *chromosomes*. Each chromosome is made up of *genes* — the 'coded' instructions for the appearance and constituents of an organism.

Chromosomes: thread-like structures containing the hereditary information found in the nucleus of the cell. They consist of DNA. They are paired — one from each parent. Each chromosome contains information for a number of characteristics, such as hair colour, eye colour, shape of ear, etc.

Chromosomes are present at all times in the nuclei of all cells. Each chromosome is made of a single molecule of *DNA* (deoxyribonucleic acid) which consists of a chain of many connected genes. Every species has its own number of chromosomes per cell, arranged in pairs. Humans have 46 chromosomes in body cells and 23 in sex cells.

Note that sex cells have half the number of chromosomes that are present in other cells. Gametes are produced by a type of cell division called *meiosis*, in which 'daughter' cells receive exactly half the number of chromosomes as the original nucleus. When non-sex cells (somatic cells) divide for growth or repair, the daughter cells receive the same number of chromosomes as the original by a type of cell division called *mitosis*.

When sex cells come together at fertilisation the 'correct' number of chromosomes is restored. The new individual gets half of its chromosomes from each parent sex cell, so the new individual's cell has the total number of chromosomes of both parent cells combined.



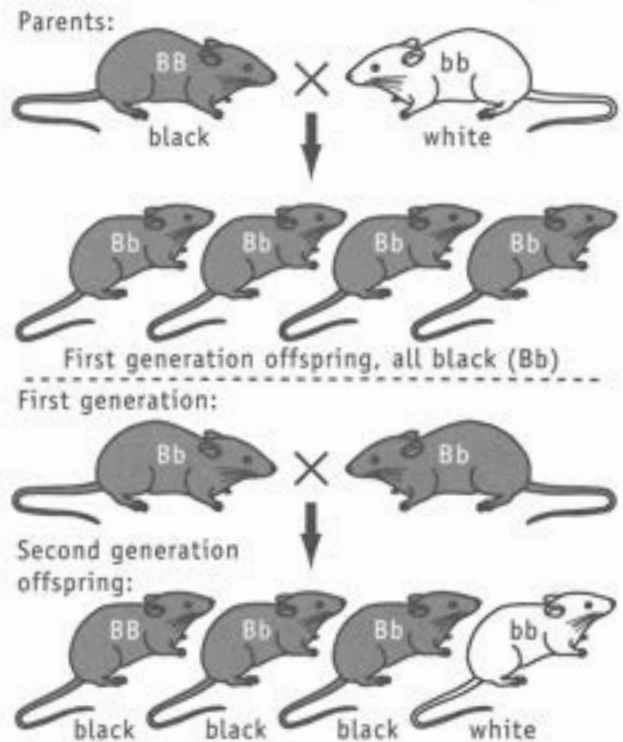
Genes are the basic unit of inheritance and relate to specific characteristics (traits) of organisms. In humans, for example, these could be blood group, hair colour, etc. These genes (with the exception of the sex chromosomes) are paired. These paired genes (one from each parent) control the same characteristic and may give identical instructions. However their instructions may also be different, in which case the instructions from the *dominant* gene will 'mask out' those from the *recessive* gene. Such non-identical genes are called *alleles*.

On this sheet, we look at an example of single-factor inheritance that follows the laws established by Mendel — the colouration of rats. In this case the allele for black is dominant over the allele for white.

If we call the allele for white *b* and the allele for black *B*, it implies that the allele with the capital letter is dominant.

If a homozygous black rat (with identical alleles *BB*) mates with a homozygous white rat (*bb*), all the offspring will be heterozygous black (*Bb*).

If these offspring are then mated, they give a mixture of black rats and white rats. The ratio of black rats to white rats will be 3 : 1.



- 1 Define 'genetics'.
- 2 Who could be said to have been the 'father' of modern genetics? What organisms did he study?
- 3 Which type of reproduction leads to genetic variation?
- 4 What is the general name for a sex cell?
- 5 What do we call a fertilised egg?
- 6 State the number of chromosomes present in human body cells.
- 7 State the number of chromosomes present in human sex cells.
- 8 List the difference between 'mitosis' and 'meiosis'.
- 9 Briefly explain what is meant by 'allele'.
- 10 Briefly explain what is meant by 'homozygous'.
- 11 Briefly explain what is meant by 'heterozygous'.
- 12 'Sexual reproduction gives offspring that are identical to the parents.' True or false?
- 13 'Chromosomes are found in the nucleus of a cell.' True or false?
- 14 'Genes are the basic units of inheritance.' True or false?
- 15 'The term "dominant" refers to the characteristic that can always be seen in the offspring, for example brown eye colour is dominant over blue eye colour.' True or false?

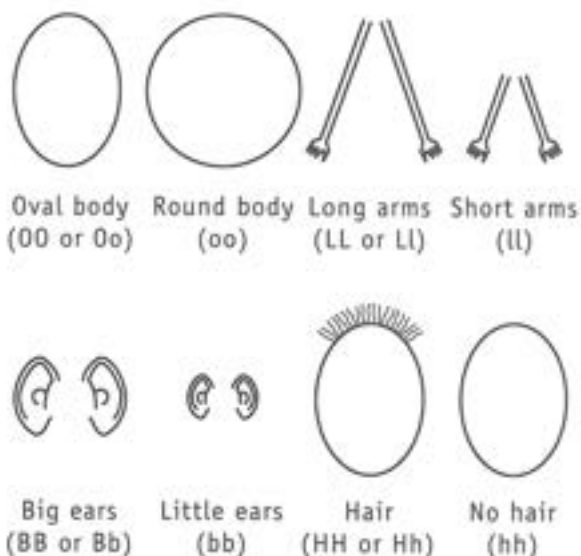


Genotypes and Phenotypes

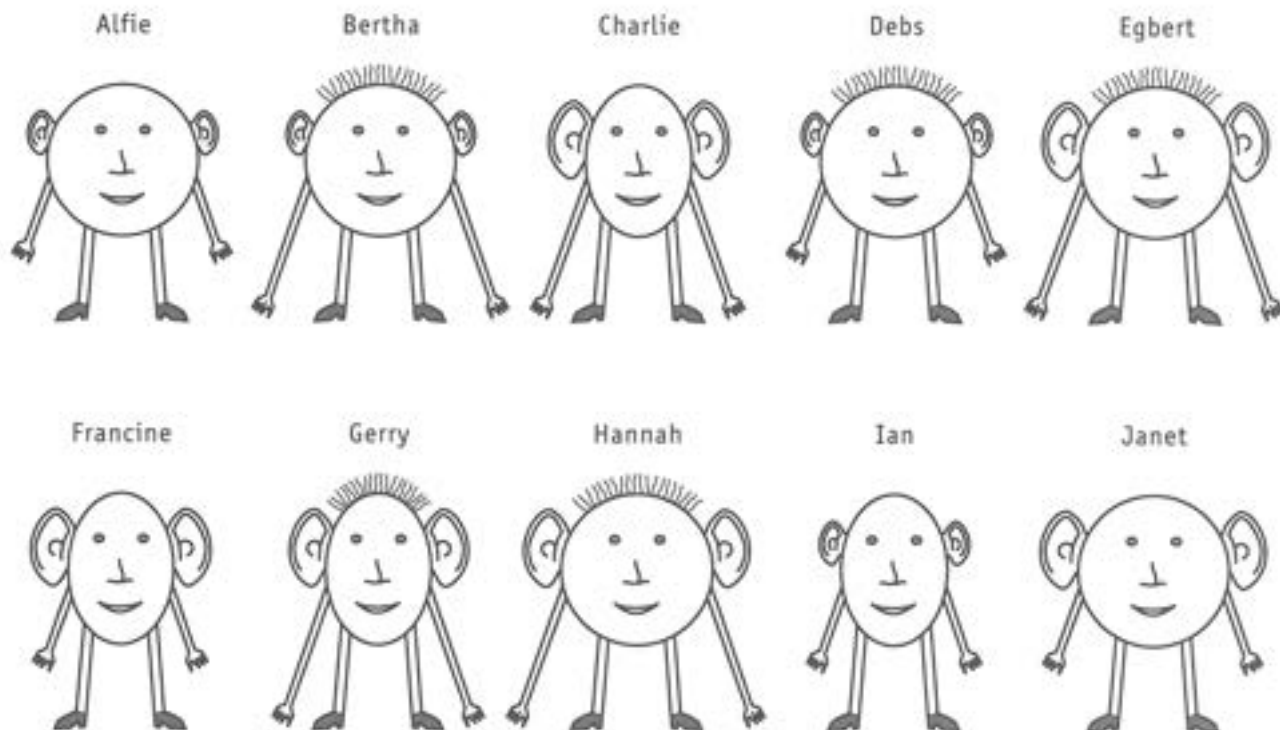
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The Griblets are a race of beings who live on the planet Gribbo. They look strange, but their genetics follow the same pattern as those of humans on Earth. We are going to study four of their physical characteristics (phenotypes). These are shape of body, length of arms, size of ears and hair. The genotypes (genetic composition) and phenotypes are shown on the right.

- 1 What is meant when the genotypes are shown in upper case (for example H) or lower case (for example h) letters?



The pictures below show a 'family album' of a group of Griblets.



- 2 Draw up a table, showing the name, and the phenotypes and genotypes for body, arms, ears and hair for each Griblet.
- 3 Which Griblets have identical phenotypes?
- 4 Do the individuals mentioned in Question 3 have the same genotypes for the traits illustrated? Explain your answer.
- 5 What conditions would be needed for the seemingly identical Griblets to be identical twins?
- 6 Think of another physical characteristic of the Griblets that hasn't been mentioned. Describe two possibilities for this phenotype and your suggested genotypes for this characteristic.



What is Genetics?: the concept of heredity

7

Choose the correct word from this list to fill each space below. Each word has been used once:

<i>Adapt</i>	<i>cell</i>	<i>classical</i>	<i>colour</i>	<i>disease</i>	<i>DNA</i>
<i>evolution</i>	<i>expressed</i>	<i>genes</i>	<i>genetics</i>	<i>genome</i>	<i>genotype</i>
<i>half</i>	<i>inherited</i>	<i>Mendel</i>	<i>molecules</i>	<i>origins</i>	<i>pea</i>
<i>pedigree</i>	<i>phenotype</i>	<i>populations</i>	<i>selective</i>		

How often have you been told that you resemble a parent (and groaned as a result)? We now understand that many features are _____ from our parents and, in turn, from their parents and so on, as a result of each parent contributing _____ of their offspring's genetic make-up. Resemblances occur when one parent's contribution is _____ more strongly than the other parent's. These features can be as simple as eye or hair _____, while other features can have more serious consequences, such as the inheritance of _____.

It is important to understand the difference between the terms *phenotype* and *genotype*. _____ is used to describe observable features, such as hair colour, etc. The genetic make-up of an individual is its _____. This includes all genes inherited from both parents, **whether or not they are expressed**. Genetic material, _____, is stored in the nucleus of a _____.

There are three main branches of genetics: *Mendelian*, *molecular* and *population*. Mendelian genetics is also known as _____ genetics because it is the branch begun over a century ago by Gregor Mendel. _____ studied how certain features are transmitted from one individual, in the case of his studies the _____ plant, to its offspring. This form of _____ breeding is still used today when we want to produce, or enhance, a certain characteristic in a plant or animal. A common example is the selective breeding of farm animals for sale as meat, or for *purebred* (_____) stock. Because it is the study of transmission, Mendelian _____ is often called transmission genetics.

As the name suggests, molecular genetics studies genetics at the level of individual _____. Molecular geneticists study the composition of _____, how they are controlled and what they produce when they are all put together. The complete set of genetic information an organism contains is called its _____. The ability to produce genetic 'maps' is a result of advances made in molecular genetics.

Population genetics examines _____ as a whole, rather than individual organisms. Population geneticists might ask questions such as, 'What makes one group better able to _____ than another population of the same species?', 'What are the differences and similarities among populations?' and 'How has _____ directed the way populations have changed over time?' Population genetics reveals relationships between species, often providing clues to the _____ of species.



Match each term with its correct definition/description by writing the letter corresponding to the description in the blank box next to the word it describes.

	Term			Description
1	allele		a	Name for the normal number of chromosomes in a (non-sex) cell.
2	asexual		b	Variations on a gene for a particular characteristic.
3	cell		c	Genetic make-up of an individual.
4	characteristic		d	Progeny (children) of a plant or animal.
5	chromosome		e	Reproduction involving male and female.
6	diploid		f	A group of similar organisms capable of interbreeding.
7	dominant		g	Thread-like structure consisting of DNA, found in a cell's nucleus.
8	gamete		h	Characteristic masked by a dominant one.
9	generation		i	Name for cells only having half the 'normal' number of chromosomes.
10	genetics		j	Male sex cell.
11	genotype		k	Coworker with Crick, involved in DNA research.
12	haploid		l	The result of a sperm fertilising an ovum.
13	heterozygous		m	Offspring of the same set of parents.
14	homozygous		n	Process by which organisms produce offspring.
15	meiosis		o	Having the same alleles for a particular characteristic.
16	Mendel		p	Sudden change in DNA causing genetic variation.
17	mitosis		q	Reproductive germ or sex cell.
18	mutation		r	How sex cells reproduce.
19	offspring		s	How most cells reproduce.
20	recessive		t	His laws of inheritance are the basis for modern genetics.
21	reproduction		u	Study of heredity or passing on of characteristics.
22	sexual		v	Feature passed on to offspring, such as eye colour.
23	species		w	Having two kinds of allele for a particular characteristic.
24	sperm		x	The characteristic always seen in the offspring.
25	Watson		y	Basic unit of all living creatures.
26	zygote		z	Reproduction that doesn't involve sex.



Self-assessment: 'genetics'

9A

On these sheets there are 35 statements on 'genetics'. In the boxes provided write **T** if you think the statement is true, **F** if you think it is false, or **U** if you are unsure, or don't think you have covered that topic. After a class discussion, go over your responses, using the spaces to make any relevant comments.

- | | |
|---|--------------------------|
| 1 Genes are short lengths of chromosomes. | ☐ |
| 2 Chromosomes are found in the nucleus of a cell. | ☐ |
| 3 Meiosis and mitosis are types of cell division. | ☐ |
| 4 Deoxyribonucleic acid (DNA), has a very simple structure. | ☐ |
| 5 Mutations are very common. | ☐ |
| 6 Another name for gametes is sex cells. | ☐ |
| 7 In organisms which reproduce sexually, there are two copies of every gene. | ☐ |
| 8 Alleles are alternative forms of a gene. | ☐ |
| 9 If an individual has identical alleles, it is heterozygous for that gene. | ☐ |
| 10 The entire set of alleles of an individual is known as its genotype. | ☐ |
| 11 Alleles may be dominant or recessive. | ☐ |
| 12 The phenotype is the physical characteristics of an organism. | ☐ |
| 13 A 'monohybrid cross' is the inheritance of many characteristics. | ☐ |
| 14 Darwin investigated inheritance of characteristics in sweet peas. | ☐ |
| 15 The first generation of offspring are called F_2 in genetics. | ☐ |
| 16 Cystic fibrosis and sickle-cell anaemia are recessive genetic diseases. | ☐ |
| 17 Female sex cells are made up of two X chromosomes. | ☐ |



Self-assessment: 'genetics'

9B

- 18 A child's gender is determined by the type of chromosome passed on by the father.
- 19 Human cells, apart from sex cells, usually have 46 chromosomes.
- 20 Women can carry a 'haemophilia' gene without themselves being affected.
- 21 A species is a group of very similar organisms that are capable of interbreeding.
- 22 When a sperm fertilises an ovum, the result is a zygote.
- 23 Cells only having half the 'normal' number of chromosomes are diploid.
- 24 Meiosis is how sex cells reproduce.
- 25 Cloning is a type of sexual reproduction.
- 26 Offspring receive 50% of their genetic information from each parent.
- 27 All mutations are harmful.
- 28 Because *Drosophila* (fruit flies) breed quickly they are used in genetics experiments.
- 29 Offspring is another name for children.
- 30 Female sex cells are called ova.
- 31 Organisms are not affected by the environment in which they live.
- 32 Cells which have the 'normal' number of chromosomes are haploid.
- 33 Mutations can be caused by mistakes in copying DNA.
- 34 Punnett squares show the different genotypes and phenotypes of offspring.
- 35 Asexual reproduction results in genetically identical offspring.

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Total = _____

I was unsure about questions _____.



Cell division plays an essential role in the growth, repair and reproduction of multicellular organisms. Cell division is the splitting of one (parent) cell into two identical (daughter) cells. There are two main types of cell division — *mitosis* and *meiosis*. Cells fall into two main classes *somatic* cells and *germ* cells.



Somatic cells: cells making up the tissues, organs, and parts of the body other than the germ (sex) cells.

Somatic cells (somatic, from the Greek, meaning 'body') are involved in growth and body maintenance. Mitosis (binary fission) produces new cells for growing organisms and also replaces cells, which have been damaged or have died. Many single-celled organisms reproduce *asexually* (without sex) using mitosis. Before cells divide in mitosis they grow, otherwise they would be smaller after each cell division.

Diploid: double, refers to two of each chromosome, a complete set. (Remember **Diploid: Double.**)

Somatic cells are usually *diploid* ($2n$), meaning they have a full set of chromosomes, one from each parent.

Germ cells: also called gametes, for example egg (ovum) cells or sperm cells.

Gametes (from the Greek, meaning 'wife') are set aside for reproductive purposes, but are not always used. These cells undergo *meiosis*, a cell division that produces *haploid* (n) cells. These cells require fusion (joining) with gametes from the opposite sex to restore the complement (full number).

Haploid: half, refers to half of one full set of chromosomes. (Remember **Haploid: Half.**)

Human females are born with a lifetime's supply of gametes. When they reach maturity, one gamete (sometimes more) will be released monthly. Human males are not born with their gametes. Instead, males begin to form gametes at maturity and they are then produced as required.

The older a woman is when she conceives, the greater the risk there is of her child suffering a genetic disorder. The risk becomes significant above the age of 35. For fathers, the risk of producing defective gametes also increases with age, but only after the age of 50 does this pose a problem.

- 1 What type of cell division produces cells used in the growth and repair of the body?
- 2 What type of cell division produces sperm and ova?
- 3 Give another name for a 'gamete'.
- 4 What happens to the number of chromosomes in mitosis?
- 5 Why do cells undergoing mitosis need to grow before they divide?
- 6 What is the difference between 'diploid' cells and 'haploid' cells?
- 7 What makes the gametes of a mature female more likely to be defective than the gametes of a man of the same age?
- 8 'All living things grow by cell division.' True or false?
- 9 'Binary fission means two cells joining.' True or false?
- 10 'Ovulation is the process by which females release eggs every month.' True or false?
- 11 'Human males are born with a complete set of gametes.' True or false?
- 12 'Daughter cells are always female.' True or false?

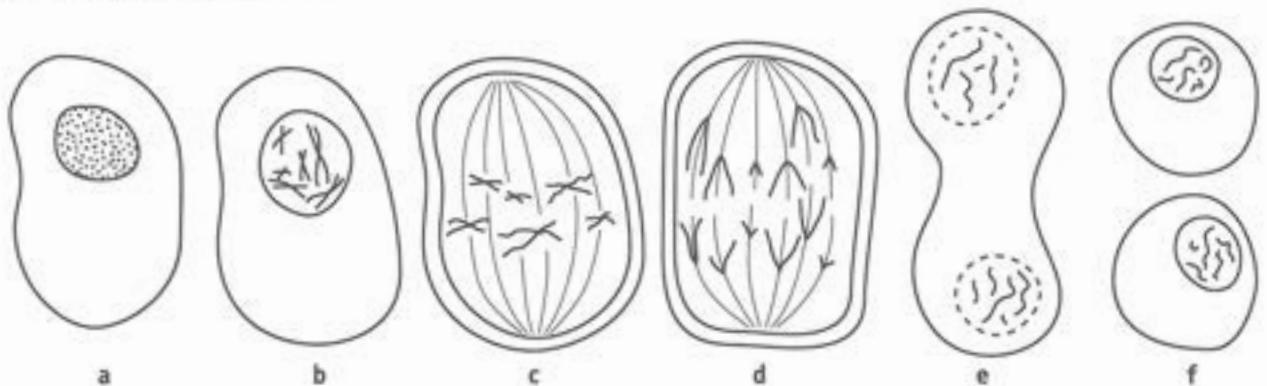


When human sex cells (gametes) join at fertilisation they produce a single cell (a *zygote*). This cell starts to divide and, after nine months (and billions of cell divisions), a baby is born.

There are thin, thread-like, microscopic structures called chromosomes present in the nucleus of a cell. The chromosomes can only be seen when a cell is dividing. Chromosomes occur in cells in pairs (apart from in sex cells). Humans have 23 pairs ($2n$) of chromosomes. Other species have different numbers of chromosomes. Chromosomes are strands of DNA (deoxyribonucleic acid), and consist of sections called genes. These contain the instructions for controlling cells, and need to be able to be copied precisely, because a mistake made early in replication will be repeated in every cell.

There are two processes by which cells can divide: *mitosis* and *meiosis*. In mitosis the cell continues to duplicate and divide, producing daughter cells that are genetically identical to the parent cell. These are needed for growth and to replace the millions of cells that die every day.

1 Mitosis occurs in *somatic* cells. What are 'somatic' cells?



This diagram shows a simplified version of the sequence followed in mitosis (nucleus is enlarged for clarity).

- a The parent cell is resting, no chromosomes are visible.
- b Pairs of duplicated chromosomes, 'sister chromatids', become visible. (**prophase**)
- c The sister chromatids line up along the centre ('equator') of the cell. (**metaphase**)
- d The sister chromatids split and move to opposite poles. (**anaphase**)
- e Two new daughter nuclei formed. Cell splits into two. (**telophase**)
- f Each new (daughter) cell has the same number of chromosomes as the parent cell.

Mitosis occurs in most body cells. It also happens in organisms that reproduce asexually.

Hint: Mitosis produces twice the numbers of chromosomes as meiosis. Remember this by **twice** and **mitosis**. Or, remember that the first three letters of mitosis are '**mit**', the same as '**mitten**', and there are two mittens in a pair.

2 What is meant by 'asexual reproduction'?



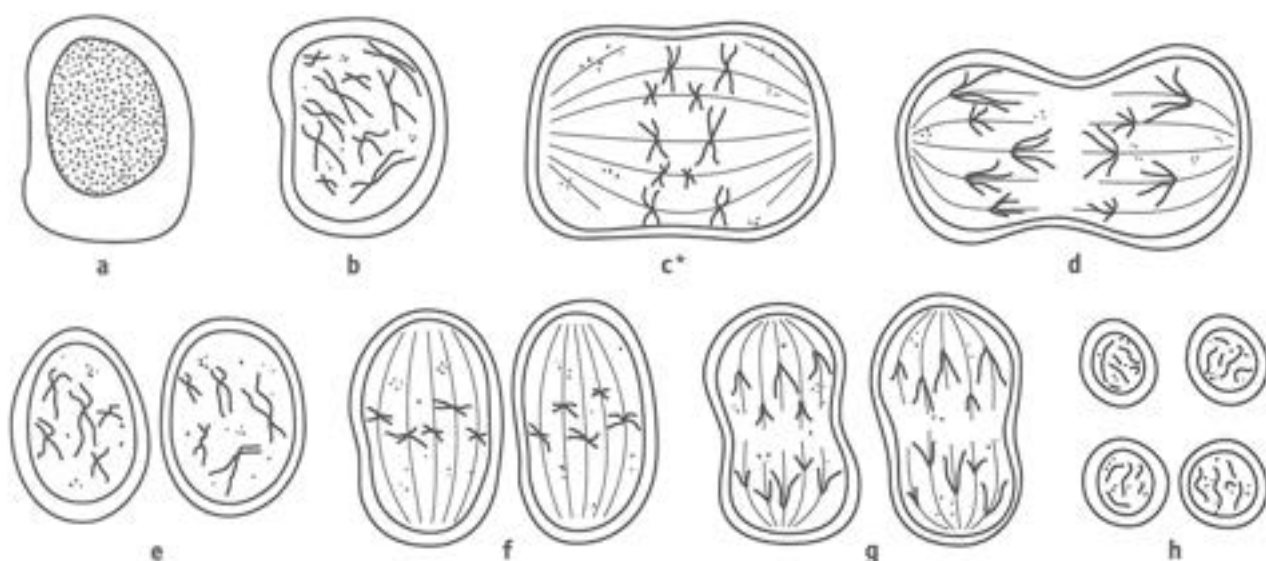
Cell Division II

11B

Meiosis occurs when sex (germ) cells are produced. If the sex cells in humans each contained the full number of chromosomes, then the zygote would have 46 pairs. The number of chromosomes present in a 'normal' body (somatic) cell is called the *diploid* number. Meiosis produces four daughter cells, each of these having half the number of chromosomes (n) of the original parent cell (that is the *haploid* number).

Meiosis (from the Greek, meaning 'to make smaller') is more complex than mitosis. Both processes involve four phases — prophase, metaphase, anaphase and telophase — but meiosis involves extra processes. For this reason, the phases (a) to (e) below are labelled prophase I, metaphase I and so on, and from (e) to (h) they are labelled prophase II, metaphase II, etc.

The diagram and steps listed below describe a simplified sequence for meiosis (nucleus enlarged).



- Parent cell (no chromosomes visible).
- Chromosomes become visible due to 'thickening'.
- Sister chromatid pairs pair up and align at 'equator'. (*Note: pairs of sister chromatids from *both* parents are present.)
- Chromatid pairs divide for the first time.
- Cell divides for the first time.
- Sister chromatids align for the second time.
- Second cell division.
- Four daughter cells formed, each with half the original number of chromosomes, all genetically different.

Meiosis helps cause genetic variation. One stage causes genes to mix. This helps ensure that new organisms are never identical to their parents. Humans can produce over 8 million types of sperm or ova, and there are 70 000 000 possible ways that these gametes can join.

- Describe (with the aid of diagrams) what happens in the prophase.
- Describe (with the aid of diagrams) what happens in the metaphase.
- Describe (with the aid of diagrams) what happens in the anaphase.
- Describe (with the aid of diagrams) what happens in the telophase.
- Give one reason why meiosis only produces half (n) the chromosomes as mitosis ($2n$).
- Assuming the population of the world is around 5 billion, why is it very unlikely to have two unrelated people with the same genetic make up?
- Name two major differences between mitosis and meiosis.



Reproduction: the creation of new life.

Reproduction occurs in all living things. The two main types of reproduction are *asexual* and *sexual* reproduction. Asexual (without sex) reproduction occurs in simple organisms. It only requires one parent and produces offspring that are genetically identical to that parent.

Sexual reproduction is carried out by all *angiosperms* (flowering plants) and most animals. This process involves the *fusion* (joining) of a male *gamete* and a female *gamete* (gametes are also called sex cells). When these cells join, *fertilisation* takes place. The result is a *zygote*, the first cell of a new living thing. Sexual reproduction produces offspring which are not genetically identical to either parent.

In animals, male gametes are referred to as *spermatozoa* (often shortened to *sperm*) and female gametes are referred to as *ova* (singular *ovum*). Human sperm are about 0.05 mm in diameter, and ova are about 0.2 mm in diameter.

Gametogenesis: the process by which gametes are produced and matured.

Female gametes, in general, are large because they have a food reserve and are *non-motile* (they don't tend to move). Male gametes are much smaller, and are motile.

Gametes have only half the number of chromosomes (they are *haploid*) compared to the organism that produced them. They are produced by *meiosis*, a type of cell division (see worksheets 10 and 11, 'Cell Division I' and 'Cell Division II'). When the gametes fuse, the *zygote* has the original number of chromosomes of the parent organism (it is *diploid*).

Meiosis produces gametes that vary genetically because, in a process called 'crossing-over', genetic material is exchanged between chromosomes. Because of this, each gamete has its own unique characteristics. A *zygote* will *resemble* its parents, but will show differences.

Most animals are unisexual — males produce only sperm and females produce only ova. Most plants combine male and female roles, they can be defined as *hermaphrodites*. These produce both male and female gametes. (Some definitions of 'hermaphrodites' include only animals, not plants.)

Certain simple invertebrates can reproduce by *parthenogenesis*, in which eggs develop without being fertilised by a male sex cell.

Before an organism can release gametes it must be sexually mature. The age of maturity can range, from a few weeks in the case of small mammals (such as mice and rats), to years in larger mammals (approximately 13 years for humans and 18 years for elephants).

- 1 Give definitions for all of the terms in *italic* type in the passage above.
- 2 'Asexual reproduction occurs in complex organisms.' True or false?
- 3 'Zygotes produced by sexual reproduction will genetically resemble the parents.' True or false?
- 4 'Large mammals reach sexual maturity earlier than small mammals.' True or false?
- 5 'Sperm are generally larger than ova.' True or false?
- 6 'Invertebrates are animals without backbones.' True or false?





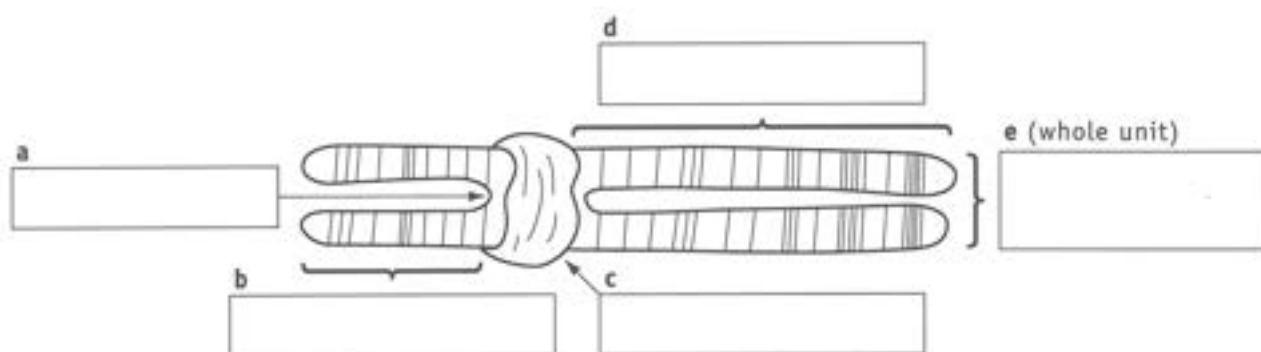
DNA: various nucleic acids that are the basis of heredity and are localised in the nuclei of cells.

Genetic material is found in the form of chromosomes, made of DNA (deoxyribonucleic acid) and proteins. DNA has a backbone with a tightly-coiled, long double-helix structure (resembling a spiral staircase). The material of chromosomes is called 'chromatin', which means 'coloured threads'. This is what genetic material resembles after it has been stained and viewed under a microscope.

Chromatids: pairs of parallel strands of a duplicated chromosome joined by a single centromere.

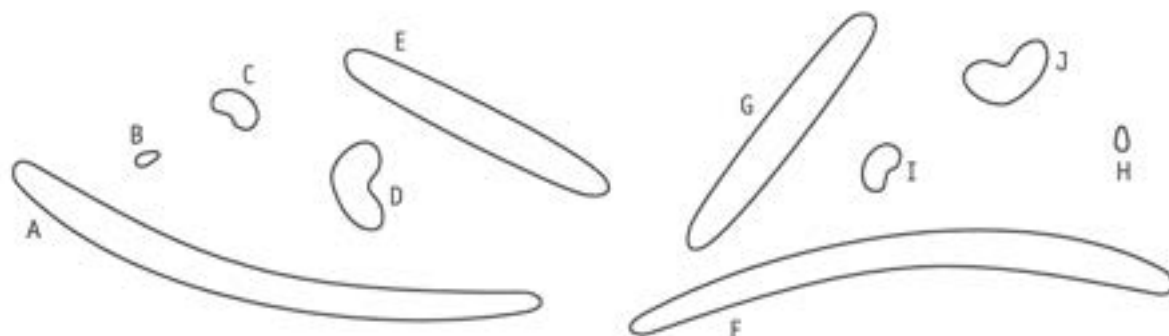
When genetic material replicates, the two copies remain attached for a time. While in this state, each thread-like structure is called a *chromatid*. The pair are referred to as *sister chromatids*. This difference in naming tells us that the genetic material is still undergoing replication. The chromatids are attached at one area called the *centromere*. A protein 'belt' (a *kinetochore*) forms around the centromere. Each part of the sister chromatids is referred to as an arm. The short arms are called *p*-arms (from the French, *petit* meaning 'small') and the long arms are *q*-arms.

1 Reread the text above and label the diagram below.



Chromosomes tend to stay separate most of the time but, during meiosis, this changes. In humans each of the 23 chromosomes contributed by the mother has a matching chromosome contributed by the father, giving a total of 23 pairs. These matching pairs are called *homologous pairs*, or *homologues*.

2 Use five different colours (one per pair) to show which of the chromosomes in the diagram below belong together.



3 What is the shape of a DNA molecule?

4 What is a 'chromatid'?



Crossing over exchanges genetic information. It occurs when the chromosomes in homologous pairs cross over each other, breaking pieces off, which they then exchange with each other. At this point, the chromosomes are in their duplicated state, so many different combinations can occur.

- 1** Carry out the following exercise, working in small groups.

Step 1: Decide on two different colours for each of the homologous pairs shown at the bottom of the page (for example red/white for one pair and blue/yellow for another). Agree on a sequence for how your group will colour the four sets. Each student in the group should end up with homologous groups that look alike.

Step 2: Working alone, cut out your homologous pairs. Align them and randomly cut part of one arm away. Swap the pieces around, rejoining the genes from one (for example red/white) with the other (for example blue/yellow). Repeat the process several times, including all four arms, then return to your group and compare crossovers.

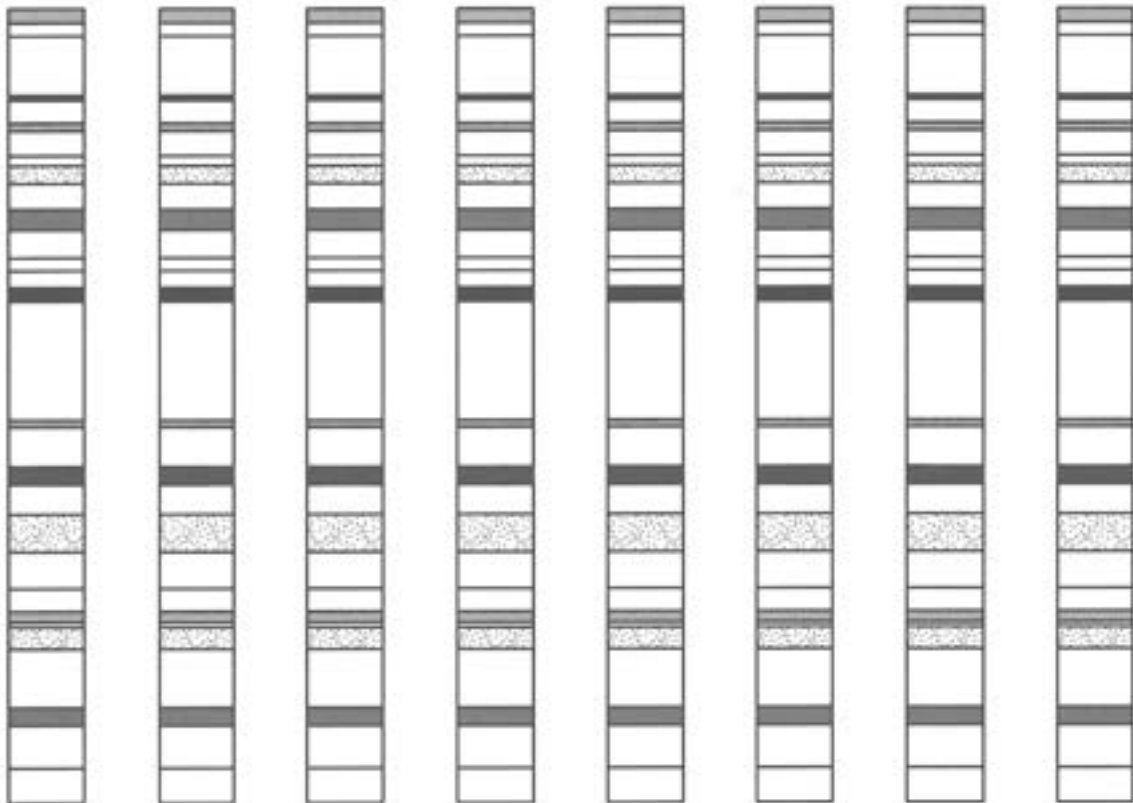
- 2** Did all crossovers produce identical outcomes?

- 3** Explain your answer to Question 2.

- 4** Look at the results of other students. What do they tell you about 'gene-shuffling'?

In reality, crossovers occur again and again, sometimes returning part of the genetic material to the original chromatid.

- 5** Why do you think that some family members resemble each other more closely than others?





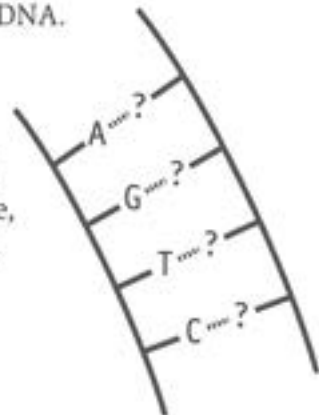
DNA: the molecular basis of heredity, found in cell nuclei, consisting of a double helix held together by weak bonds between bases which project inward from two chains.

Compared to other molecules, DNA is huge and amazing. However, it is made up of only a few substances: four nitrogen 'bases', a sugar and phosphates. The four bases are adenine (A), thymine (T), cytosine (C) and guanine (G). A base, a sugar and a phosphate form a three-part unit called a nucleotide. The sugar involved is deoxyribose, hence the name deoxyribonucleic acid, or DNA.

Base-pairing

In the same way that jigsaw puzzles fit together, nucleotides have specific pairing rules. Adenine on one strand will pair with thymine on the other, and guanine on one strand will pair with cytosine, producing a ladder-like structure, pulled together by hydrogen bonds into a coil shape, known as a *double helix*.

The diagram on the right shows a short sequence on a DNA 'ladder'.



1 Copy the diagram, using the pairing rules to identify the bases labelled '?'.
2 What do the dotted lines between the bases represent?

Although the amount and proportions of the nitrogen bases vary in all species, the four bases are all present and always follow the base pairing rules: A with T, and G with C.

Hint: think *Ginger Cake*. *Ginger* and *Cakes* go well together, and so do *G* and *C*. If you remember this, that leaves the pairing of A and T remaining by default.

The genetic code can be compared to a four letter alphabet: A, G, C and T. These bases spell the genetic 'words', and these groups of words are known as genes. The language of DNA is a language of proteins. Proteins are made of *polypeptide* chains.

Replication

When DNA replicates (produces a copy of itself), it unwinds and unzips first. The bonds that held the complementary bases together break and each half of the 'zipper' is used as a *template* (model) on which new DNA will be built.

RNA

Like DNA, RNA (ribonucleic acid) is a nucleic acid and controls some of the cell's chemical activities. RNA is formed using DNA as a template and having three of the same bases as DNA. The exception is that instead of thymine it uses uracil (U) to bond with adenine. Combinations of the four 'letters' A, G, C and U then 'spell' the names of amino acids that form the polypeptide chains. For example, the base sequence GCU is the amino acid, alanine. Each 'triplet' of bases is called a *codon*. As the spelling (base sequence) is 'read out' (three letters at a time), the amino acids called for are linked together as a polypeptide chain. Once this is complete the RNA winds itself into specific shapes, depending on the order in which they were assembled, to form organelles, enzymes, and other structures.

- 3 Give the full names of the bases A, G, C, T and U.
- 4 Give the full names for DNA and RNA.
- 5 What substances make up a 'nucleotide'?
- 6 Complete this sentence: Proteins are made of _____ chains.



Correctly complete the definitions below (1–8) using these words:

bases

DNA

homologues

hydrogen

nucleotide

polypeptide

replication

template

- 1 A huge molecule made up of repeating units of four nucleotides. _____
- 2 The production of new strands of DNA. _____
- 3 A three-part unit made of a nitrogen compound, a sugar and a phosphate. _____
- 4 An abbreviation for nitrogen-based substances found in DNA. _____
- 5 The model on which a new strand of DNA is built. _____
- 6 Corresponding chromosomes in a cell. _____
- 7 Proteins are made of chains of these. _____
- 8 Weak bonds holding strands of DNA in a double helix. _____

Quick thinking!

- 9 You may have seen the symbols ♂ and ♀ used in genetics. What does each of these symbols signify?
♂ means _____ ♀ means _____
- 10 Research where the symbols referred to in Question 9 originated.

- 11 One strand of DNA replicates a total of six times. How many strands will result?

Alleles

The word 'allele' comes from the Greek, meaning 'of one form and another'. Alleles are the alternate forms of a gene. They are genes occupying the same position on homologous chromosomes. For example eye colour is determined by a gene, but that gene has several different alleles: brown, blue, green, grey and, occasionally, violet. If you consider genes as cake mixes on sale in a supermarket, the alleles would be the different flavours of cake mix available. All the mixes (alleles) would produce a cake, but the one you buy and take home, will determine what eventually comes out of the oven. In a similar manner, you will inherit an eye colour allele from each of your parents and these will pair up to form an eye colour gene. The two alleles you inherit will determine your eye colour.

Consider a child whose mother has blue eyes and whose father has brown eyes.

- 12 Will the child have one brown eye and one blue eye? _____
- 13 Will the child have an eye colour that is a blend of brown and blue? _____
- 14 What eye colour might the child have? Explain your answer. _____

- 15 Describe another characteristic that is inherited from parents on an 'either/or' basis. _____
- 16 Describe a characteristic that is inherited from parents on a 'blend/mixture' basis. _____



When a gene for a characteristic can appear in two different forms, these are called alleles.

The basic unit of inheritance is the *gene*. When genes are available in alternative forms, these alternative forms are called *alleles*. The genes on paired chromosomes control the same characteristic and can give identical or conflicting instructions to the cells. If the instructions are different, only one gene will be obeyed. The instructions from one gene will override the other.

The gene that 'wins' is called the *dominant* gene and the one that 'loses' is called the *recessive* gene. Dominant characteristics will always appear in the first generation offspring (F_1). When a dominant trait (allele) is present, it will always mask the presence of a recessive characteristic (allele). The dominant allele is represented by an upper-case (capital) letter and the recessive allele by a lower-case letter. Recessive characteristics only become visible in their 'pure' (*homozygous*) form, when a double dose of that recessive gene is present. Since we receive one gene from each parent, the only way for the recessive to be visible in the *phenotype* (physical characteristic) is if it is the only type of gene passed on by the parents.

In Mendel's experiments he crossed peas producing green seeds with peas producing yellow seeds. 'G' represents an allele for yellow seeds and 'g' represents an allele for green seeds. Study the Punnett square on the right, which shows a cross between two heterozygous parent plants. (See worksheets 21 and 22 about Punnett squares.) It is called a monohybrid cross because it only involves one characteristic.

	G	g
G		
g		

- 1 Complete the Punnett square.
- 2 What will be the ratios of the phenotypes of the F_1 offspring?

3 Give the genotypes of a homozygous green plant.

4 Give the genotypes of a homozygous yellow plant.

Partial dominance

Sometimes a cross does not result in clear-cut differences between parents and offspring. Characteristics sometimes appear to blend in the F_1 generation. The 'Four O'Clock Plant' can occur in three colours: red, pink and white. The alleles present are R and r. A red plant will be RR, a white plant will be rr and a pink plant will be Rr. Neither characteristic is dominant.

- 5 Complete the Punnett square on the right, showing the results of a cross between a red plant and a white plant.
- 6 What will be the ratios of the phenotypes of the F_1 offspring?



Dominant Alleles

17B

- 7 Complete the Punnett square on the right, showing the results of a cross between two pink plants.
- 8 What will be the ratios of the phenotypes of the offspring of two pink plants?

- 9 Is the 'blending' occurring in the genotypes or in the phenotypes? Explain your answer.

Codominance

Short-horn cattle can be (among other colours) red-haired or white-haired. If a red-haired animal is crossed with a white-haired animal, the resulting offspring are not completely red, white or pink. They have *both* red and white hairs and are said to be *roan* coloured. The alleles for red and white both produce an effect when they are present together. We represent codominant alleles as both having capital letters, for example R_1 = red, and R_2 = white.

- 10 Complete the Punnett square on the right, showing the results of a cross between two roan cattle.

- 11 What will be the ratios of the phenotypes of the offspring of two roan cattle?

- 12 Complete the Punnett square on the right, showing the results of a cross between a roan cow and a white bull.
- 13 What will be the ratios of the phenotypes of the offspring of a roan cow and a white bull?

One example of codominance in humans is the inheritance of some blood groups.

- 14 How many characteristics are involved in a 'dihybrid cross'? _____
- 15 What organism was used by Gregor Mendel in his studies of inheritance? _____
- 16 What is another name for a 'pure-breeding' organism? _____
- 17 What is meant by F_1 ? _____
- 18 'Homozygous means having identical alleles for a characteristic.' True or false? _____
- 19 'A phenotype is a physical characteristic.' True or false? _____



You may need to use a text book or dictionary to help you complete this sheet.

Genotype and phenotype: which is which?

- 1 Write the meaning of 'genotype'. _____
- 2 Write the meaning of 'phenotype'. _____

More definitions

- 3 Match the given meanings with the correct word or prefix.

▪ zygote* ▪ homo ▪ dominant ▪ hetero ▪ gamete ▪ recessive	▪ from the Greek word meaning 'other', different ▪ standing back, subordinate ▪ mature sex cells ▪ cell formed by the union of two gametes ▪ ruling or controlling ▪ from the Greek word meaning 'same', alike.
--	--

(*Note: 'zygous' is often found in words referring to the zygote.)

Putting the pieces together: definitions

- 4 Complete the table, using the information provided in the box below.

homozygous dominant	aa	dominant	heterozygous
dominant	AA	homozygous recessive	Aa
			recessive

Definition	Correct term from choices given	Trait expressed (phenotype)
Two identical alleles of a dominant trait.		
One of each allele, dominant and recessive.		
Two identical alleles of a recessive trait.		

Recognising symbols

- 5 Group the allele pairs given below according to the descriptions:

- homozygous dominant _____
- heterozygous dominant _____
- homozygous recessive _____

Allele pairs:	Qq	rr	Rr	AA	RR	bb	Aa	QQ	Bb	qq	aa	BB
---------------	----	----	----	----	----	----	----	----	----	----	----	----



- 1 The location at which an allele is found on a chromosome.
- 2 To have a double dose of the same allele.
- 3 Two famous names in genetics are Hardy and .
- 4 The name for each duplicated chromosome, while they remain attached.
- 5 Protein 'belt' found around centromeres.
- 6 Term used to describe how genes are passed from generation to generation.
- 7 A gene that can mask the presence of another by overriding it, is said to be: .
- 8 A method of cell division producing genetically different daughter cells.
- 9 Correct term used to describe making copies of chromosomes.
- 10 An alternative form of a gene.
- 11 Means 'to have half' of a chromosome set.
- 12 These are not germ (sex) cells, but develop into tissue, etc.

13 _____





Genetics Word Search

20

In the grid, find and circle the letters of each of the theme words listed below. The words are arranged vertically, horizontally, or diagonally, and can be written in reverse. When you have found all the words, you will be left with five unused letters. Write those letters in order, starting from the top left, in the answer spaces provided, to find the mystery word.

Clue: 'In genetics, it's very important to do this properly!' _ _ _ _ _

C	E	N	T	R	O	M	E	R	E	E
E	H	E	S	N	E	D	N	O	C	Q
D	E	R	N	E	P	A	Z	K	L	U
I	L	A	O	O	E	A	Y	N	D	A
T	I	R	A	M	L	D	M	I	F	T
O	X	E	E	E	O	C	E	L	L	O
E	L	D	N	I	P	S	U	G	A	R
L	O	E	B	O	N	D	O	R	H	O
C	G	S	I	S	O	T	I	M	T	U
U	R	A	C	I	L	A	I	R	E	N
N	G	B	N	S	P	I	R	A	L	D

Choose from:

ARM

BASE

BOND

CELL

CENTROMERE

CHROMOSOME

CLONE

CODE

CONDENSE

DIE

ENZYME

EQUATOR

GENE

HALF

HELIX

LETHAL

LINK

MAP

MEIOSIS

MITOSIS

NUCLEOTIDE

PAIR

PEA

POLE

RARE

ROUND

SEED

SPINDLE

SPIRAL

SUGAR

URACIL



Punnett Squares

21

Punnett squares (named after R. C. Punnett, an English geneticist, 1875–1967) are used in genetics to calculate the frequencies of the different genotypes and phenotypes among the offspring of a cross. Punnett used poultry and sweet peas to discovered many genetic linkages.

- 1 Give a brief definition of 'genotypes'.
- 2 Give a brief definition of 'phenotypes'.

We represent the dominant of the two alleles with an upper-case letter and the recessive allele with a lower-case letter. The male (δ) parent is shown on the upper part of the Punnett square and the female (φ) parent is shown on the side.

TT is the genotype for homozygous tall plants and tt is the genotype for homozygous small plants. We are going to see what happens if we cross these.

A 'Punnett square' can be used, which shows that all of the first generation offspring (F_1) will be heterozygous (Tt) plants. Because tall is dominant (in this case) over short, all of these plants will be tall.

If these offspring plants are then crossed, the next generation (F_2) will be TT, Tt and tt, in the ratio of 1 : 2 : 1. One quarter will be homozygous tall (TT), one half will be heterozygous tall (Tt) and one quarter will be homozygous small (tt).

A male German Shepherd with erect ears (EE) was mated with a female German Shepherd with floppy ears (ee).

- 3 Draw a Punnett square in the box provided on the right, showing the possible outcomes of the offspring of this cross.
- 4 Complete this summary table for the German Shepherd cross described above:

Offspring	Genotype	Phenotype
a		
b		
c		
d		

	T	T
t	Tt	Tt
t	Tt	Tt

	T	t
T	TT	Tt
t	Tt	tt

German Shepherd Cross

- 5 Summarise the rules for representing dominant and recessive alleles in Punnett squares.
- 6 What is meant by 'genetic linkage'?
- 7 What is meant by the ' F_2 generation'?



Recessive traits will be expressed if they are the only genes present. If they are linked to a dominant gene, then the dominant gene will be expressed. This is why Mendel's green peas only reappeared after a second generation. Recessive genes need time to come together again.

When recessive traits are crossed, the ratio of the recessive trait to the dominant trait in the offspring (F_1) is 1 : 3. This is illustrated by the Punnett square.

It's in the eyes!

Let's consider eye colour. In humans, brown eyes result from a dominant allele and blue eyes result from a recessive allele. Since each parent contributes an eye colour allele, a child might inherit one allele for blue eyes and one allele for brown eyes. To have blue eyes, offspring must have a double dose of recessive alleles, otherwise the dominant allele will be expressed.

- 1 Construct a Punnett square showing the possible outcomes of a cross between a brown-eyed (BB) father and a blue-eyed (bb) mother. (Remember: the female parent is shown on the side of the square and the male parent is shown on the upper part of the square.)

- 2 What will be the ratio of brown eyes to blue eyes?

_____ : _____

- 3 What will each offspring inherit in terms of eye colour alleles? _____

Consider what would happen if one of the offspring of the previous cross married a blue-eyed person. No-one can predict with certainty if any particular one of *their* children (the F_2 generation) will be blue-eyed or brown-eyed. However, the *probabilities* can be calculated.

- 4 Construct a Punnett square showing the possible outcomes of a cross between a female from the F_1 generation (Bb) and a blue-eyed male (bb).

- 5 What will be the ratio of brown eyes to blue eyes?

_____ : _____

- 6 The next generation would be called the _____ generation.

- 7 In a totally different family, two brown-eyed parents produced a blue-eyed child. What does this tell us about the genotypes of the parents?

Time to think!

Gregor Mendel used the characteristics of peas to study inheritance. Most people might think that peas are boring and it might be more interesting (and useful!) to study humans.

- 8 Give one reason why Mendel didn't study inheritance in humans.



Variations in a Species

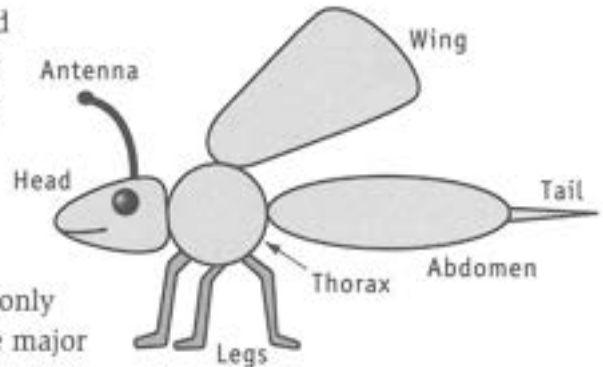
23A

You will need scissors, glue and a die to do this exercise.

Some entomologists studied a species of insect only found on Gribble Island. These unique insects seemed to have many differences, but were definitely of the same species because they could breed with each other.

The diagram shows a typical specimen found on Gribble Island.

After a lot of research it was discovered that there were only two phenotypes (physical characteristics) for each of the major body parts of the insects.



You can simulate the range of characteristics of these insects by following these instructions:

- collect sheet 23B showing possible body parts ('Insect parts'), glue, scissors and a die.
- roll the die to randomly select the characteristics of a body part. Use the table below to make your selection, depending on whether the throw of the die results in an even or odd number. The first roll of the die will select the type of head, the second will select the antenna, etc. Some characteristics (marked *) will require two throws of the die.
- assemble the insect parts, and glue the complete insect into your book. Label the parts.

Characteristic	Even Number	Odd Number
Head	Plain eye	Compound eye
Antenna	Long antenna	Short antenna
Thorax	Striped thorax	Plain thorax
Legs	Long legs	Short legs
Wings*	Large wings	Small wings
	Checkered wings	Plain wings
Abdomen*	Thick abdomen	Thin abdomen
	Dark abdomen	Light abdomen
Tail	Straight tail	Curly tail

Further Activities

- 1 After completing your insect, compare it with others made in the class.
- 2 Explain why it is unlikely that any two people in the class will have made identical creatures.
- 3 A scientist found a group of insects on another island that resembled those from Gribble Island, but did not breed with them. Are these insects of the same species as the ones you have been studying? Explain your answer.



Insect parts



Plain eye



Compound eye



Long antenna



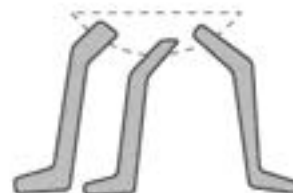
Short antenna



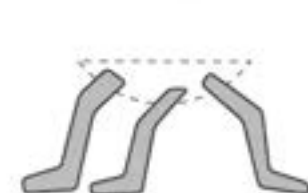
Plain thorax



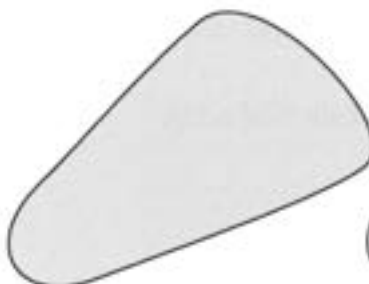
Striped thorax



Long legs



Short legs



Plain



Chequered

Large wings



Plain



Chequered

Small wings

Dark



Light



Thick abdomens

Dark



Light



Thin abdomens



Straight tail



Curly tail



Genetics is the study of inheritance — how characteristics are passed from one generation to the next. Genetic material is stored in the nucleus of each cell. The major bodies involved in this process are called chromosomes. Each chromosome is made up of genes. Genes are the instructions that are responsible for the components and appearance of an organism. They are found in the nucleus of a cell.

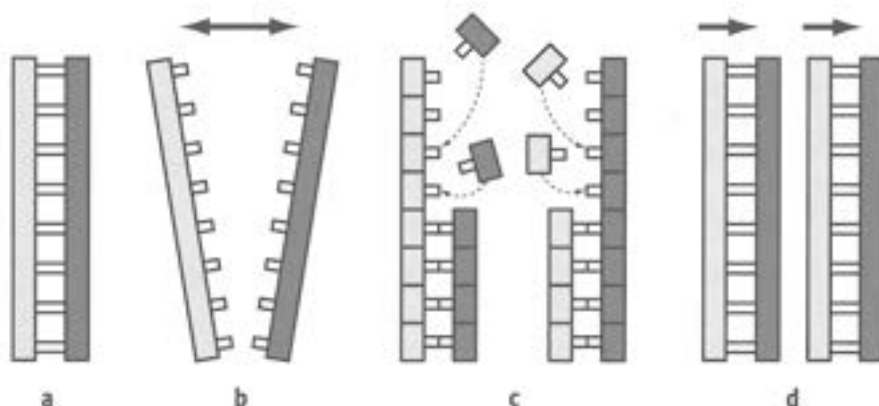
A gene: the part of a chromosome that determines the inheritance of a characteristic or of a group of characteristics.

Each chromosome is made of a single molecule of DNA. The DNA molecule is a chain of many genes connected together. DNA is a nucleic acid and is a very large molecule. If the DNA in a single cell was uncoiled and put in a straight line, it would be a metre long. The backbone of the DNA molecule is two chains wound around each other, forming a shape rather like a twisted ladder, called a double helix. There are a large number of smaller molecules (amino acids) attached. The order in which these side groups are attached determines the code of a particular nucleic acid. The code tells the cell how to reproduce itself. Cells divide in an organism for growth, repair and reproduction.

Watson, Crick and Wilkins shared the 1962 Nobel Prize for their contributions to the understanding of the genetic code. Watson and Crick came up with our current double helix (spiral) model of DNA. In humans each DNA molecule is thought to contain from a few to many thousand genes. Each gene can be thought of as a series of 'rungs' on the DNA 'ladder'.

A simplified version of DNA replication is shown.

- a The double strand of DNA.
- b The double strand 'unzips' and two single strands are formed.
- c Both strands (chromatids) are replicated (copied).
- d The strands join to form two double strands identical to the original chromosome.



DNA is made up of many nucleotides. A nucleotide is a little unit made up of a sugar, at least one phosphate and a nitrogen base. The nitrogen base contains a ring (or rings) containing nitrogen. Each half of the strand contributed by a parent contains many alleles. Alleles are particular (non-identical) forms of a gene, coding for a particular trait. The alleles can be compared to individual teeth on a zipper. When each tooth of the zipper pairs with its correct mate, the zipper is complete. Similarly, when each allele pairs up with its correct mate, a gene pair is formed.

The exact location of an allele on a strand of DNA is called its locus.

It is important that the DNA is copied exactly. Otherwise information to make new cells will be incomplete or faulty. Any mistakes will be passed on to the next generation of cells. These cells could have problems that could cause the organism to be defective or less successful in its environment.

Genes are not always copied exactly. Mutations can occur — these alter genes.

A mutation: a change in a gene (or genes). The resulting mutated gene may result in a different phenotype (physical characteristic).

Mutations can involve the sequence of genes being rearranged or changed. If these changes occur in gametes (sex cells), they can be inherited. If they are recessive (genes which do not automatically show up in physical characteristics), they may go undetected. These mutations may result in diseases, such as a blood disease called sickle-cell anaemia. Mutations can be favourable or unfavourable to a species. They lead to changes of the characteristics of individuals in the population of a group of organisms. If favourable, these changes may make those individuals more likely to survive and breed. If unfavourable, these changes may make those individuals less likely to survive and breed, so they will die out.

Mutations very rarely happen spontaneously. They are usually caused by mutagens such as:

- chemicals (acridine dyes, fungi that cause food to rot, nitrous acid, tobacco tars)
- heat (increasing temperature can cause a rise in mutation rate)
- radiation (cosmic rays, gamma rays, ultraviolet [UV] light and X-rays)

- 1 How many strands make up a DNA molecule? _____
- 2 What is the shape of a DNA molecule? _____
- 3 'A gene is a short length of DNA'. True or false? _____
- 4 Why do cells divide? _____
- 5 Why is it important that DNA is copied exactly during replication? _____

- 6 Who is credited with the current model of DNA? _____
- 7 What does the code of a nucleic acid tell a cell to do? _____
- 8 What happens in mutation? _____
- 9 Name two causes of mutation. _____
- 10 Are mutations always a disadvantage? _____
- 11 What is a locus? _____
- 12 What is a nucleotide? _____
- 13 What proportion of genetic information does a child inherit from each parent? _____
- 14 What is a 'chromatid'? _____
- 15 Give a definition of 'genetics'. _____

Syndrome: a group of characteristics occurring together that characterise a particular condition.

'Down Syndrome', also called *trisomy 21*, results from a chromosomal abnormality. It is caused by the inheritance of extra genetic material belonging to chromosome 21. *Tri* means three, and three copies of at least some of chromosome 21 are inherited. As a result the person has 47 chromosomes instead of the normal 46 in each body cell.

During meiosis, chromosome pairs usually separate completely. Occasionally this doesn't happen and part (or all) of one remains with the other. This event is called *nondisjunction* — the gamete takes away with it extra genetic information. Chromosome 21 contains information about developing facial features. A tiny flap of skin at the corner of each eye produces our eye shape. In people with Down Syndrome the extra information produces an exaggerated (epicanthic) fold resulting in a characteristic shape of the eyelids. (This can be altered by surgery.)

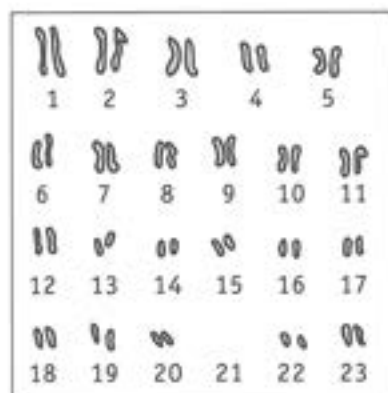
The degree of severity of the condition depends upon which extra (third) copies of genes the offspring have inherited. In general, the more extra DNA is received, the greater the effect.

A 25-year-old mother has a one in 2000 chance of having a child with Down Syndrome. This increases to one in 200 for a 35-year-old mother, and greater than 1 in 40 for women over 40 years of age. Many trisomy 21 embryos are miscarried during pregnancy, but approximately one in a thousand survive to birth.

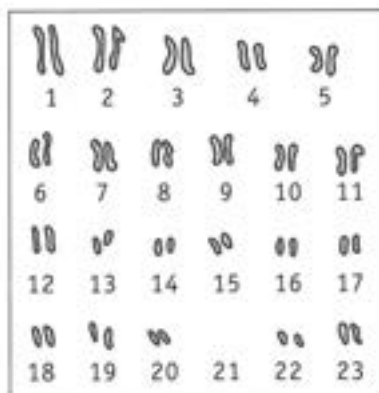
Some of the features of Down Syndrome include mild to severe mental retardation, slow body development and a small, round head. Such persons often have a cheerful disposition but are also likely to develop heart defects, and tend to develop leukemia. Until recently the average life expectancy for a person with Down Syndrome was in the teens but, nowadays, many can achieve a near average life-span.

Karyotypes are ordered displays of an individual's chromosomes. They are very useful to identify abnormalities in chromosomes. Details for chromosome 21 have been left blank below.

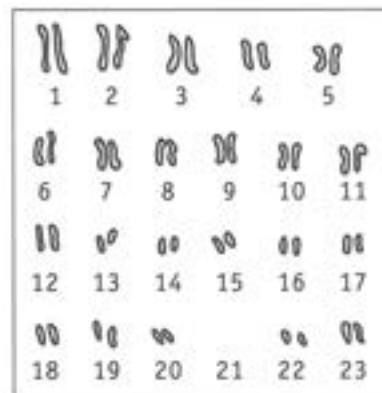
- Complete the three karyotypes below by drawing the chromosome 21 details to match the label each time.



Trisomy 21

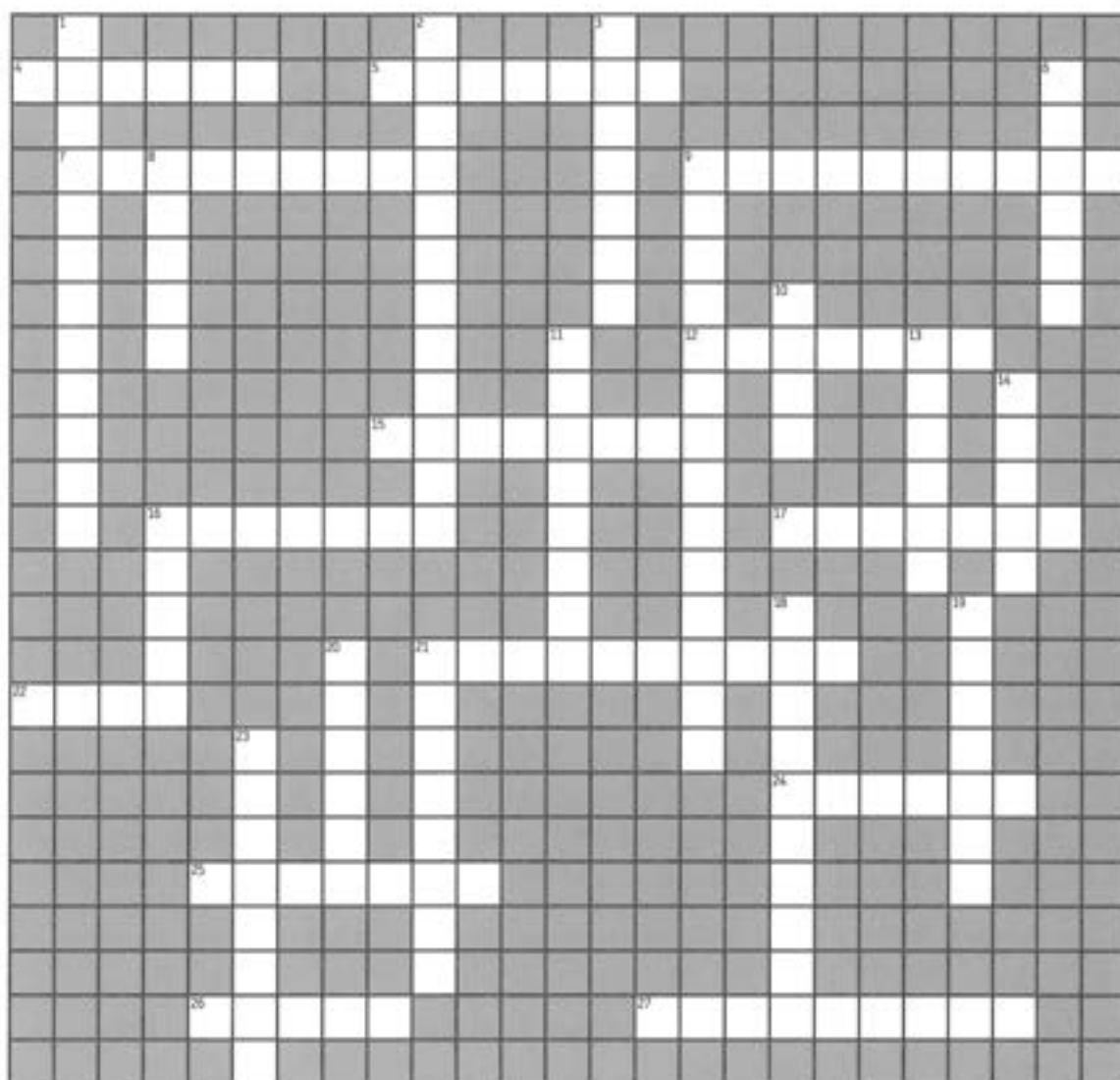


Monosomy 21



Normal

- Describe three characteristics of a child with Down Syndrome.
- What is meant by a 'syndrome'?
- Meiosis is involved in the replication of which type of cells: germ (sex) cells or somatic cells?
- Are older or younger mothers more likely to have a child with Down Syndrome?
- How many chromosomes are present in the body cells of a person with Down Syndrome?



Clues Across:

- 4 Reproduction involving male and female (6)
- 5 How sex cells reproduce (7)
- 7 Characteristic masked by a dominant one (9)
- 9 Found in nucleus, thread-like structure containing genetic material (10)
- 12 Type of reproduction not involving sex (7)
- 15 Trait that 'overpowers' the recessive (8)
- 16 Group of similar organisms capable of breeding with each other (7)
- 17 Having two of each chromosome (2n) (7)
- 21 Can be either F₁ or F₂ for example (10)
- 22 Female sex cell (4)
- 24 Cell produced by the union of two gametes (6)
- 25 Half the 'normal' number of chromosomes (7)
- 26 Where an allele is located on a chromosome (5)
- 27 Children (progeny) of a particular plant or animal (9)

Clues Down:

- 1 Process of producing offspring (12)
- 2 Having different alleles for the same characteristic (12)
- 3 How most cells reproduce (7)
- 6 Sex (germ) cell (6)
- 8 Coworker with Watson, involved in research on DNA (5)
- 9 Feature passed on to offspring (14)
- 10 Basic unit of life (4)
- 11 Genetic make-up of an individual (8)
- 13 Non-identical gene for a particular characteristic (6)
- 14 DNA's structure is a double _____ (5)
- 16 Male sex cells (5)
- 18 Having the same alleles for the same characteristic (10)
- 19 'Normal' body cell (7)
- 20 The 'Father of Modern Genetics' (6)
- 21 The study of inheritance (8)
- 23 Sudden change caused by a mistake in replicating DNA, for example (8)



Gametogenesis (a combination of the words *gameto*, meaning combined gamete, and *genesis*, meaning to be born, or originate) is the production of gametes. In order to produce viable (capable of surviving) offspring, organisms require the ability to self-replicate their genetic material. This can be summarised in three steps:

- gamete production (meiosis)
- fertilisation of the egg
- growth and development of a fertilised egg into a new organism.

Gametogenesis involves first reducing the parents' genetic material to half. When two gametes, one from each of two parents, unite the full genetic complement is restored.

Exercise

Cut out the box of text above and glue it in your notebook under the heading 'Gametogenesis'. Cut out the headings, descriptions and diagrams below. Match each phase with its description and diagram. (Note: one phase requires two illustrations). Then stick them in your book in the correct order.

TELOPHASE

PROPHASE

ANAPHASE

METAPHASE

Nuclear material thickens and pairs with its genetic homologue.

Homologous pairs (sister chromatids) are pulled apart to opposite poles.

Cytoplasm pinches off, daughter cells are formed.

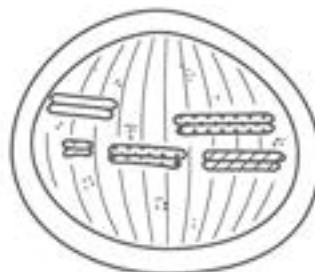
Genetic material lines up at the 'equator', spindle fibres form (to direct material to the poles).



a



b



c



d



e



Rabbits and Myxomatosis

28

In 1859, Thomas Austin of Barwon Park Estate, Western Victoria, brought a consignment of rabbits to Australia from England and released them on his property for hunting purposes. They bred slowly at first but, after three years, they had become acclimatised and began to reach pest proportions. About this time a bushfire destroyed part of the fence around Austin's property. Rabbits escaped and spread dramatically.

Six years after their introduction, Austin killed 20 000 rabbits on his own property and estimated that there were at least another 10 000 still at large. By 1879 they were in New South Wales and, within a short time, they had spread over 75% of the continent, causing extensive damage to grazing land.



Rabbits had not been a serious pest in England despite their very high birth rate. This was because, in England, they had various 'natural enemies' which preyed on them and kept the population down. These natural enemies did not exist in Australia. Because of rabbit burrows and the habit of eating roots in dry seasons, the rabbit has destroyed areas which would otherwise have regenerated with the return of favourable rainfall. In this way, one rabbit can displace many sheep and cattle from a pasture.

Large sums of money were spent on trapping, poisoning, destruction of warrens and gassing but to no avail. In theory, one pair of rabbits, in ideal conditions, *could* multiply to 13 million rabbits in three years. Finally in the early 1950s, it was decided to try biological control. Specimens of the *myxoma* virus (originally found in South America) were injected into a few rabbits which were then released. The myxoma virus is carried by fleas and mosquitoes from live rabbits, and it spread very rapidly. By 1955, 90% of the rabbit population had died of the resulting disease, *myxomatosis*.

However, in recent years, two developments have taken place. Despite the fact that the virus is widespread, the rabbit population is increasing. The virus itself seems to be becoming less deadly.

Rabbits have varied in their resistance to myxomatosis. The disease did kill 90% of the population, leaving behind 10% which were those with the strongest resistance. The new population of rabbits in Australia is on its way to becoming a new species, compared to the English rabbit which could not resist myxomatosis. The myxoma virus is spread by fleas and mosquitoes from live rabbits, not from dead ones. The more virulent strains of the virus kill rabbits more quickly and, therefore, have less chance of surviving by spreading from one live rabbit to another. This helps to explain why the virus is becoming less virulent.

Our experience with rabbits and the myxoma virus supports the theory for natural selection. Those more fit (that is more resistant to the virus) survive, those less fit, do not. Evolution pushes species to change, when there is a change in circumstances.

- 1 From what overseas country did all the rabbits in Australia originate?
- 2 Why were they brought to Australia?
- 3 Why were they a serious pest in Australia when they were not in their original country?
- 4 Fencing was tried as one way to limit the spread of the rabbit population. Why do you think that this was not entirely successful?
- 5 Why was the myxoma virus introduced into the rabbit population?
- 6 What is the difference between the rabbit population of today and that pre-1950?
- 7 What does 'highly virulent' mean?
- 8 Does the experience with rabbits and myxomatosis support the theory of natural selection? Explain your answer.
- 9 Suggest a method of controlling the rabbit population that has not been mentioned already on this sheet. List advantages and disadvantages.



The *Africanis*, a breed of dog native to Africa for over 70 centuries, is believed to be the last remaining pure breed of dog, in the truest meaning of the term 'pure bred'. It possesses features of a breed whose gene pool has been directed by natural selection alone and not by human intervention. As for all pure-bred animals, individuals share similarities. In the case of the *Africanis* these are:

- a related gene pool
- overall head and body shape
- short fur
- medium size
- friendly, intelligent and alert personality
- fitness, active behaviour and general good health.

Dogs: carnivorous mammals of the family *Canidae*. There are about 40 species, which can range from 1.5 kg to 75 kg when fully grown.

Because natural selection has acted on the *Africanis* gene pool, it remains richly varied. For this reason, although the overall appearance of the breed is recognisable, each dog has individual characteristics. These differences include different:

- colouring
- markings (solid colour or multiple colour markings).

These differences make the breed particularly attractive to humans because a choice of 'models' is available. People can select a look that most appeals to them. In order to safeguard the future of the *Africanis*, it has recently been officially recognised as a breed in the hopes it will not be interfered with or left to interbreed with other types of dog, producing mongrels.

A mongrel: an individual resulting from the interbreeding of different breeds.

Write brief definitions for the following terms with relation to their meaning in genetics.

- | | |
|---|-----------|
| 1 enhancing | 5 mongrel |
| 2 side-effects | 6 species |
| 3 native | 7 breed |
| 4 heterogenous | |
| 8 Many people in Africa consider that the <i>Africanis</i> is fitter and has a more balanced nature than many other breeds of dog. Give one reason why this breed might be 'superior' to other breeds. | |
| 9 Using a variety of sources (text books, conventional encyclopaedias, CD ROMs, the Internet, etc.) research the <i>Africanis</i> and prepare a report on it. Your teacher will tell you whether to use an oral presentation, a poster, an assignment, etc. If using the Internet, try: Kennel Union of South Africa or Africanis Society of Southern Africa. | |

Genetics and dog breeding

The term 'pure-bred' refers to an animal that is homozygous for a gene or set of genes. For example, a pair of pure-bred Irish Setters will produce a litter of similar puppies closely resembling their parents. Pure-bred dogs are bred for specific features, such as size, colour, personality, appearance, etc. Different breeds can become fashionable, simply based on these characteristics. Unwanted side-effects can sometimes be linked to features wanted by breeders. These can occur due to the enhancing of features beyond normal, healthy limits or as a result of desired features being genetically linked to a negative (unwanted) trait.



Africanis: the world's last pure breed of dog?

29B

Think about the various breeds of dogs that exist. People own dogs for various reasons, such as companionship, or to help with work, etc.

- 10 Choose three examples of dog breeds that stand out in your mind. State the image/stereotype associated with each dog. Describe the type of person that might choose each of the dogs described.

Enhancing (increasing) certain features can sometimes lead to unwanted complications. Many 'toy' breeds produce litters with very small puppies, having high mortality rates among the puppies.

- 11 Describe what is meant by a 'toy breed'. Give an example.
- 12 What is meant by the term 'high mortality rate'?
- 13 Why might some toy breed females have difficulty in producing live young?
- 14 Give three factors which contribute to the cost of pure-bred puppies.
- 15 Describe the possible positive (good) and negative (bad) effects resulting from selective breeding for the following traits:
- short, flat snout (nose)
 - chunky (compact) body, short legs
 - long body
 - loose skin, many folds
 - narrow snout
 - protective feelings for owner/property.

Selective breeding can produce unwanted features due to 'genetic linkage'. For example a certain dog breed was once popular because it was said to be safe around young children. A new colour was produced by a rarely occurring recessive mutation (unexpected genetic change). This attractive colour quickly became popular and many litters were produced. A 'brindle' variation followed, by breeding animals of the new colour with animals of standard colours.

Brindled: dark streaks or flecks on a grey or tawny background.

Over some years, and after a few regrettable incidents, people observed a link between the recessive mutation, the new colouration and aggression. This new, unexpected and unpredictable dangerous behaviour made the breed less attractive to couples with young children. A once reliable breed had become potentially dangerous, and many people are now reluctant to leave dogs of this type alone with young children.

- 16 Study the following paragraph and explain the words printed in **bold** type.

A **commercial** breeder of this type of dog said, "The breed has been **contaminated**. Some **brindle** examples of our dogs could now be considered **unreliable**. The well known original coloured **pure-bred** dog is very safe. The **mutation** has given rise to **aggressive** behaviour. But even some dogs that don't exhibit any trace of the new colour can be considered unreliable because the characteristic is **recessive**."

- 17 Prepare a report/presentation about a pure-breed of dog of your choice. Include:
- special features for which the breed is known, including a photo/illustration of an example of the breed
 - the breed's country of origin and how long the breed has been in existence
 - the reputation of the breed (for example what sort of people choose it, the type of image it projects)
 - the advantages and disadvantages that may have resulted from selective breeding in the breed.

If possible, contact a breeder for information about the breed. Compare this information during a class discussion.

Links have been made between defective pigmentation genes and vision. The gene that produces the Siamese cat's unique colouration is linked to crossed eyes!





Genes are the potential of an organism. The environment may or may not allow the full development of this potential. Genes are contained in chromosomes in the nucleus of cells. Genes are short lengths of chromosomes and are the basic unit of inheritance. They usually copy themselves exactly each time a cell divides (meiosis and mitosis) but, occasionally, the unexpected happens — and the unexpected is usually (but not always) unpleasant.



Genes are made of deoxyribonucleic acid (DNA), which has a very complicated structure. If this structure is altered in any way, a different gene may result. This is called *mutation*. Alleles are different genes that influence one characteristic. Mutations can remove one allele and replace it with another.

Mutations very rarely occur spontaneously — perhaps one in a million may happen without apparent cause. The rate is increased by substances called *mutagens*. Some of these are produced by the cell themselves and others come from the environment. These include:

- some chemicals
- radiation
- increased temperature.

Chemicals that are suspected of being mutagens include nitrous acid (HNO_2) and nitroso compounds which are present in treated meats such as salami and bacon, mustard gas (used in warfare), tobacco tars, and fungi that spoil foods. (Some of these substances may also cause cancer — they are *carcinogenic*.)

Radiation includes gamma radiation (a form of nuclear radiation), X-rays, cosmic rays (from space) and ultraviolet (UV) light. There are also natural radioactive sources from the Earth itself. The ozone layer protects us from much of the harmful radiation from space, but this has been damaged by such man-made chemicals as CFCs (chlorofluorocarbons).

An increase in temperature can cause a five-fold increase in the rate of mutations. Violent temperature changes can cause an even greater change in rate.

If mutations occur in gametes (germ or sex cells), they are inherited. They are usually recessive and may initially go undetected. They will occur in the phenotype (appearance of the individual) if the homozygous condition (containing identical alleles) occurs. The possibility of this is increased by in-breeding.

The overwhelming majority of mutations are harmful to the organisms affected, sometimes (very rarely), however, they may be beneficial (helpful).

Natural selection

This has sometimes been referred to as 'survival of the fittest'. It is a process by which the environment selects favourable variations and eliminates (gets rid of) harmful or less successful ones. (Environment means the surroundings or conditions in which the organism lives.)

After many generations of the selection and 'weeding-out' of the less fit, the characteristics of a population may change and the population may become better adapted to the current environment. If an organism is suited to its environment and there are no external pressures, the organism can remain unchanged.



Charles Darwin was a scientist who, in the middle of the 19th century, after looking at animals in South America, the Galapagos Islands and Australia, came up with a theory of evolution to account for the diversity of animal life throughout the world. His theory can be summarised by the points below.

- In a population there are variations. All the members of a particular species are not identical.
- In any generation, there are offspring that don't reach maturity (grow up) and reproduce. The characteristics of these organisms are removed from the population.
- The organisms that survive and reproduce are well adapted to that particular environment. They have favourable variations.
- Favourable variations are passed on to offspring. They become more and more common in the population.

Alfred Russel Wallace (1823–1913) was a British naturalist who also developed a theory of evolution based on natural selection, at the same time as Darwin. While working in the East Indies, Wallace noted some fundamental differences between Asian animal species and Australian species. He placed a hypothetical 'dividing line' known as *Wallace's Line* between the islands of Lombok and Bali (Indonesia), between Sulawesi and Borneo and between the Moluccas and the Philippines. Many mammals, fish and birds occur in large numbers on one side of Wallace's line but, a few kilometres on the other side of the line, they either don't occur or are in much fewer numbers.

- 1 Name the two processes involved in cell division and replication.
- 2 What is an 'allele'?
- 3 What does a 'mutagen' do? Give three specific examples of mutagens.
- 4 A 'carcinogen' causes what disease?
- 5 Give a definition of 'gamete'.
- 6 Give a definition of 'phenotype'.
- 7 Give a definition of 'mature'.
- 8 Are all mutations harmful? Explain your answer.
- 9 Summarise the four points in Darwin's 'Theory of Evolution'.
- 10 Why might animals like giraffes develop from relatively short-necked animals over a period of time when grazing is in short supply?
- 11 The crocodile has remained almost unchanged over 100 million years. Why might this be so?
- 12 What is the source of 'cosmic' rays?
- 13 What does this statement define? *'A permanent change in hereditary material that involves either a physical change in chromosome relations or a biochemical change in genes.'*

Research

- 14 Photocopy or draw a map showing Australia and neighbouring Asian neighbours. Draw in Wallace's Line.



Questions 1 to 15 are multiple choice.

Choose the response which best answers the question each time.

- 1** A gamete is a:
A sex cell **B** mutation **C** characteristic **D** nucleus
- 2** A zygote is:
A a sex cell **B** pollen **C** a fertilised sex cell **D** a type of DNA
- 3** A chromosome is:
A something in the nucleus of a cell **B** the coloured part of a cell
C part of a gene **D** a blood group
- 4** A gene is:
A a chromosome **B** all the chromosomes in a cell
C part of a chromosome **D** always dominant
- 5** How many chromosomes are in each normal human body cell?
A 23 **B** 26 **C** 43 **D** 46
- 6** Which of these represents the sex chromosomes of a female?
A XX **B** XY **C** YY **D** AO
- 7** The fusion (joining) of sperm and ovum is called:
A fertilisation **B** determination **C** generation **D** mutation
- 8** In the study of inheritance, special terms are used to describe the genetic structure and appearance of an individual. An organism which has identical genes for a particular characteristic would be said to have a:
A homozygous phenotype **B** heterozygous phenotype
C homozygous genotype **D** heterozygous genotype
- 9** In mice, the gene for a rough coat (R) is dominant over the recessive gene for a smooth coat (r). A rough coat male was crossed with a smooth coat female. If the male was Rr then the proportion of smooth coats to rough coats in the offspring would be:
A 2 : 2 **B** 1 : 3 **C** 3 : 1 **D** 4 : 0
- 10** The genes that determine if a mouse is an albino come:
A from the male parent only **B** from the female parent only
C one from each parent **D** two from each parent
- 11** A plant with a red flower is crossed with one with a white flower. All of the offspring have pink flowers. If two pink flowers were then crossed, the offspring from these would be:
A all pink **B** red or pink **C** pink or white **D** red, pink or white
- 12** The situation in Question 11 was an example of:
A dominance **B** recessiveness
C recession **D** incomplete dominance



- 13 A pea plant which is homozygous for tallness is crossed with one which is homozygous for shortness. Given that the gene for tallness is dominant over that for shortness, the percentage of heterozygous tall plants in the first generation of offspring should be:
- A 100% B 75% C 50% D 25%
- 14 The possibilities for heredity and evolutionary changes are greatest in species which reproduce by:
- A fusion B budding C sexual means D spores
- 15 The physical appearance of a particular characteristic is the:
- A genotype B phenotype C gender D pedigree

Questions 16 to 18 refer to the following information:

In mice there are 40 chromosomes in each body cell. The sex of a mouse is determined by the same number of chromosomes as in humans.

- 16 How many chromosomes does a mouse receive from each parent?
- 17 How many non-sex chromosomes are normally found in a body cell from a female mouse?
- 18 How many non-sex chromosomes are present in a mouse gamete?
- 19 *Drosophila* are often used in genetics experiments. Give another name for *drosophila*.
- 20 Why are *Drosophila* used in genetics experiments?
- 21 What are 'alleles'?
- 22 Cells can be damaged by outside causes, which can cause mutations. Give two possible causes of mutations.
- 23 Some diseases, such as muscular dystrophy, can be passed on from generation to generation by a sex-linked mechanism. Name two other diseases that can be inherited in this way.
- 24 Who was the Austrian Monk who first investigated inheritance in peas?
- 25 The type of cell division which produces gametes is _____.
- 26 The type of cell division which results in two cells identical to the parent cell is _____.

For Questions 27 to 31, answer true or false.

- 27 'In most countries, including Australia, the number of females born compared to the number of males is approximately in the ratio of 105 females to 100 males.' True or false?
- 28 'The genetic make-up of an individual comes equally from its father and its mother.' True or false?
- 29 'The first generation of individuals produced from a set of parents is sometimes called F_2 in genetics.' True or false?
- 30 'Cloning is a type of asexual reproduction.' True or false?
- 31 'Punnett squares are a way of working out possible combinations of genes.' True or false?



Answers

1 Darwin and Evolution I

- 1 A rock or mineral, usually from another time in geological history, containing the imprint of a plant or animal.
- 2 True.
- 3 A quick growing grass would do better because it could be replaced in less time.
- 4 Factors may include cow's preferences, rainfall (some species require less water than others), growth rate, nutrients available.
- 5 Pressure applied on a species, either by nature or humans, that directs the evolution of a species.
- 6 Examples include: Natural — weather, food availability, availability of territory (such as nesting sites). Human — removal/introduction of species and/or chemicals, modification of habitat.
- 7 Answers will vary. Teacher to check.
- 8 Answers will vary. Teacher to check.
- 9 The surroundings (habitat) in which a species lives.

2 Darwin and Evolution II

- 1
 - Applies directional pressure because it does not allow the plant to establish. Eventually all seed reserves will be used up and the species will become extinct unless reintroduced.
 - Does not apply pressure. The species is not naturally occurring and completes its lifecycle within a year. Unless reintroduced it will not return.
 - Applies directional pressure by encouraging/eliminating the growth of species in the area.
- 2 This question is designed to encourage logical thought. Students should recognise that, over time, the long-beaked species will increase in number while the original phenotype decreases. This is the result of long-beaked species being more successful at obtaining mudworms, subsequently, their growth and reproduction will be better than that of the short-beaked birds.
- 3 We do not know if the parent was homozygous (AA) or heterozygous (Aa) for the gene. We do not know which sexes are carriers, if it is a sex-linked trait (though we can assume it is not sex-linked).
- 4 If homozygous AA, then more offspring will be born with the long-beaked trait and directional evolution will occur more quickly than for Aa. If the gene is sex-linked, evolution will occur at different rates for each sex.
- 5 The effects of genes on an individual can affect whether or not it survives, but populations illustrate the "big picture" of changes occurring. With reference to the birds in the exercise, short-beaked birds may survive and reproduce but the trend toward a shift in beak shape/size is an example of population evolution.
- 6 A specific characteristic observed.
- 7 The progeny/child.
- 8 Teacher to check individual responses. The answer must describe how the species benefits in competition with others.

3 The Theory of Evolution

- 1 Teacher to check individual responses.
- 2 Answers will vary. Exception may be taken to the disregard of a Creator (that is God) and the mention of a scientific explanation for evolution.
- 3 young/offspring.
- 4 similar (identical).
- 5 survive.
- 6 traits/characteristics.
- 7 increase.
- 8 die.
- 9 Generally, the closer a species is to the primary producer, such as those that provide food for many species (insects, frogs, etc.), the more offspring are produced. Also, larger animals require more energy to produce their larger offspring, therefore fewer are born.

- 10 To ensure that at least some individuals survive to prevent the species from becoming extinct.
- 11 Based on a theory or idea.
- 12 To do with the current living times of whatever is being discussed (that is the species in question).

4 Gregor Mendel: 'The Father of Genetics'

- 1 A person who studies plant life.
- 2 Does not require another individual to fertilise.
- 3 Easily grown, grow quickly, clearly identifiable characteristics to observe, able to self-fertilise, can be eaten.
- 4 Yellow/green colour (also wrinkled/smooth seed coat).
- 5 Yellow is dominant as illustrated by the F₂ generation all being yellow.
- 6 To increase the accuracy of his calculations (to eliminate exceptions from skewing results).

5 Genetics: an overview

- 1 The study of the inheritance of an individual's characteristics.
- 2 Gregor Mendel (1822–84). He studied the pea plant.
- 3 Sexual.
- 4 Gamete (germ cell).
- 5 Zygote.
- 6 46
- 7 23
- 8 Mitosis: cell division used to produce body cells (somatic) genetically identical to parent, undergoes one series of divisions. Meiosis: cell division produces genetically different sex cells (gametes), undergoes two divisions.
- 9 Alternative form of a gene.
- 10 Double dose of the same allele for a given trait, for example AA or aa.
- 11 A pair of different alleles for a trait, for example Aa.
- 12 False.
- 13 True.
- 14 True.
- 15 True.

6 Genotypes and Phenotypes

- 1 Upper case dominant and lower case recessive.
- 2

Name	Body	Arms	Ears	Hair
Alfie	oo round	ll short	bb little	hh hairless
Bertha	oo round	LL/LI long	bb little	HH/Hh hair
Charlie	OO/Oo oval	LL/LI long	BB/Bb big	hh hairless
Debs	oo round	ll short	bb little	HH/Hh hair
Egbert	oo round	LL/LI long	BB/Bb big	HH/Hh hair
Francine	OO/Oo oval	ll short	BB/Bb big	hh hairless
Gerry	OO/Oo oval	LL/LI long	BB/Bb big	HH/Hh hair
Hannah	oo round	LL/LI long	BB/Bb big	HH/Hh hair
Ian	OO/Oo oval	ll short	bb little	hh hairless
Janet	oo round	ll short	BB/Bb big	hh hairless

- 3 Egbert and Hannah.
- 4 Possibly, if both alleles are the same for each genotype, but this may not be so i.e. ears could be Bb for one and BB for other.
- 5 Same sex, and same alleles for all traits.
- 6 Possibilities include length of leg, size of head, etc. Select an appropriate letter for the characteristic (one not already used) and employ appropriate use of upper-case and lower-case letters.

7 What is Genetics?: the concept of heredity

Correct order is: inherited, half, expressed, colour, disease, phenotype, genotype, DNA, cell, classical, Mendel, pea, selective, pedigree, genetics, molecules, genes, genome, populations, adapt, evolution, origins



Answers

8 Genetic Terms

1 b, 2 z, 3 y, 4 v, 5 g, 6 a, 7 x, 8 q, 9 m, 10 u, 11 c, 12 i, 13 w, 14 o, 15 r, 16 t, 17 s, 18 p, 19 d, 20 h, 21 n, 22 e, 23 f, 24 j, 25 k, 26 l.

9 Self-assessment: 'genetics'

1 T, 2 T, 3 T, 4 F, 5 F, 6 T, 7 T, 8 T, 9 F, 10 T, 11 T, 12 T, 13 F, 14 F, 15 F, 16 T, 17 T, 18 T, 19 T, 20 T, 21 T, 22 T, 23 F, 24 T, 25 F, 26 T, 27 F, 28 T, 29 T, 30 T, 31 F, 32 F, 33 T, 34 T, 35 T.

10 Cell Division I

- 1 Mitosis.
- 2 Meiosis.
- 3 Sex cell.
- 4 They are duplicated and the daughter cell contains the same amount as the parent cell.
- 5 To accommodate extra nuclear material, cytoplasm, etc. for division.
- 6 Diploid contain a full chromosome complement, haploid contain only half.
- 7 The fact that they are older, and have been exposed to radiation (such as X-rays, computers), medication, etc. Male gametes are produced as needed and are not exposed to these things for years at a time.
- 8 True.
- 9 False.
- 10 True.
- 11 False.
- 12 False.

11 Cell Division II

- 1 Body cells.
- 2 Non-sexual reproduction, cloning.
- 3 Chromosomes condense and are visible in the nucleus of the cell.
- 4 Sister chromatids move toward the 'equator' of the cell to align.
- 5 Sister chromatids split apart, move to opposite poles.
- 6 Final stage, nuclear membrane pinches off, daughter cells formed.
- 7 When fusion occurs the full set is re-established. If the set was not halved, after fusion there would be twice as many chromosomes as required.
- 8 Individuals are all unique and gene shuffling produces billions of possible combinations.
- 9 Mitosis produces identical daughter cells to the parent, it undergoes only one series of divisions and is used to produce somatic cells. Mitosis produces genetically different daughter cells, undergoes two divisions and is used to produce gametes.

12 Gametes and Reproduction

- 1 asexual (reproduce without sex), sexual (reproduce sexually), angiosperms (flowering plants), fusion (joining) gamete (sex cell), fertilisation (fusion of male and female gametes), zygote (first cell produced by fertilisation), spermatozoa or sperm (male gametes), ovum or ova (female gametes), non-motile (do not move), haploid (half set of chromosomes), meiosis (cell division used for gamete production), diploid (full set of chromosomes), crossing over (exchange of genetic material between chromosomes), resemble (look like), hermaphrodites (combining male/female role), parthenogenesis (egg development without fertilisation)
- 2 False.
- 3 True.
- 4 False.
- 5 False.
- 6 True.

13 Chromosomes I

- 1 a centromere b p-arms c kinetochore
d q-arms e sister chromatids
- 2 Pairs are: AF, BH, CI, DJ, EG.
- 3 spiral, double helix.
- 4 A duplicated chromosome in cell division (it has a twin 'sister').

14 Chromosomes II

- 2 No.
- 3 Combinations will vary depending on how students cut and pasted.
- 4 When many thousands are being shuffled, variety is bound to occur.
- 5 Because genes from the same parents are being shuffled, reducing the possible options.

15 DNA I

- 1 AT, GC, TA, CG.
- 2 Hydrogen bonds.
- 3 Adenine, guanine, cytosine, thymine and uracil.
- 4 DNA deoxyribonucleic acid, RNA ribonucleic acid.
- 5 A base, a sugar and a phosphate.
- 6 Polypeptide.

16 DNA II

- 1 DNA.
- 2 Replication.
- 3 Nucleotide.
- 4 Bases.
- 5 Template.
- 6 Homologues.
- 7 Polypeptide.
- 8 Hydrogen.
- 9 Male, female.
- 10 male: spear and shield of Roman god Mars, female: hand mirror of Roman goddess Venus.
- 11 128.
- 12 No.
- 13 No.
- 14 Child will have brown if father is homozygous (BB) dominant, may be brown or blue if he is heterozygous (Bb) depending on which allele he passes to his child. Mother can only provide blue (bb genotype) so the child will be either Bb (brown) or bb (blue).
- 15 For example hair colour, or gender.
- 16 For example skin colour, or eye shape.

17 Dominant Alleles

	G	g
G	GG (yellow)	Gg (yellow)
g	Gg (yellow)	gg (green)

- 2 yellow : green = 3 : 1
- 3 gg
- 4 GG

	R	r
R	Rr (pink)	Rr (pink)
r	Rr (pink)	rr (pink)

- 6 All pink (Rr). (pink : white : red = 4 : 0 : 0)



Answers

7

	R	r
R	RR (red)	Rr (pink)
r	Rr (pink)	rr (white)

- 8 Pink : white : red = 2 : 1 : 1
9 Blending in phenotype (appearance), alleles remain unchanged.

10

	R ₁	R ₂
R ₁	R ₁ R ₁ (red)	R ₁ R ₂ (roan)
R ₂	R ₂ R ₁ (roan)	R ₂ R ₂ (white)

- 11 Red : roan : white = 1 : 2 : 1

12

	R ₁	R ₂
R ₁	R ₁ R ₂ (roan)	R ₂ R ₂ (roan)
R ₂	R ₂ R ₁ (white)	R ₂ R ₂ (white)

- 13 Red : roan : white = 0 : 2 : 2
14 Two.
15 Pea plants.
16 Pedigree and homozygous.
17 First filial, first generation offspring.
18 True.
19 True.

18 Understanding Genetic Terms I

- Genetic make-up of an individual.
- Outward appearance of an individual as a result of how its genes are expressed.
- Zygote: cell formed by the union of two gametes
 - Homo: from the Greek word meaning 'same', alike
 - Dominant: ruling or controlling
 - Hetero: from the Greek word meaning 'other', different
 - Gamete: mature sex cell
 - Recessive: standing back, subordinate
- Two identical alleles of a dominant trait: homozygous dominant, AA/dominant.
One of each allele, dominant and recessive: heterozygous, Aa/dominant.
Two identical alleles of a recessive trait: homozygous recessive, aa/recessive.
- Homozygous dominant: AA, RR, QQ, BB.
Heterozygous dominant: Aa, Qq, Rr, Bb.
Homozygous recessive: rr, bb, qq, aa.

19 Understanding Genetic Terms II

- 1 locus, 2 homozygous, 3 Weinberg, 4 chromatid, 5 kinetochore, 6 inheritance, 7 dominant, 8 meiosis, 9 replication, 10 allele, 11 haploid, 12 somatic 13 Answer: 'Oh brother, not more peas!'

20 Genetics Word Search

'In genetics, it's very important to do this properly': ALIGN

21 Punnett Squares

- An organism's genetic makeup, whether or not genes are expressed in appearance.
- An organism's appearance based on which genes are expressed.

3

	E	e
E	EE (erect)	Ee (erect)
e	Ee (erect)	ee (erect)

- All genotypes are Ee and all phenotypes are erect ears.
- Dominant alleles are represented by capital letters, recessive alleles are represented by lower-case letters.
- The relationship between, often seemingly unrelated, genes where one affects the expression of another.
- Second filial, generation ('grandchildren' of an original set of parents).

22 More Punnett Squares

1

	B	b
B	BB (brown)	Bb (brown)
b	Bb (brown)	bb (brown)

- 2 Brown eyes : blue eyes = 4 : 0
3 Each will inherit a brown dominant allele and a blue recessive allele.

4

	b	b
B	Bb (brown)	Bb (brown)
b	bb (blue)	bb (blue)

- 5 Brown eyes : blue eyes = 2 : 2
6 F₂
7 Both parents were heterozygous Bb, each gave a blue allele.
8 The life span of humans is too long, not enough individuals would offer themselves as samples, no way of knowing if obvious traits are the only available alleles, for example a recessive disease could be hidden.

23 Variations in a Species

- Teacher to check all parts are complete.
- There are many possibilities/combinations.
- No, members of the same species can breed with each other.

24 DNA: the raw material for genetics

- Two.
- Spiral, double helix.
- True.
- To reproduce, replace and repair.
- So errors are not duplicated.
- Watson, Crick and Wilkins.
- How to assemble protein molecules.
- The original DNA sequence is altered.
- Chemicals, heat, radiation, or spontaneous changes.
- Usually (less than 1% are beneficial).
- The site of a gene on a strand of DNA.
- A section of DNA containing a sugar, a base and a phosphate group.
- Half from each.
- A duplicated strand of DNA, attached to its 'sister' at the centromere.
- The study of inheritance.

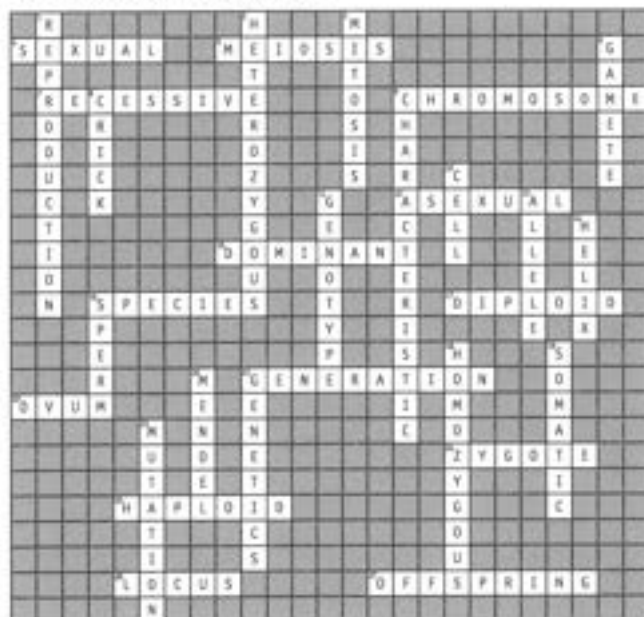
25 Genetic Disorders: Down Syndrome

- Trisomy 21: 3 copies (000), Monosomy 21: 1 copy (0), Normal: 2 copies (00).
- Mild to severe retardation; slow body movement; small, round head; heart defects; potential for leukaemia, etc.
- A recognisable group of symptoms occurring together in an individual.
- Germ (sex) cells.
- Older mothers.
- 47 (extra material from chromosome 21).



Answers

26 Genetics Crossword



27 Gamete Production: gametogenesis

Corresponding illustration given in brackets. Prophase (e): Nuclear material thickens and pairs with its genetic homologue. Metaphase (c): Genetic material lines up at the 'equator', spindle fibres form (to direct material to the poles). Anaphase (b then d): Homologous pairs (sister chromatids) are pulled apart to opposite poles. Telophase (a): Cytoplasm pinches off, daughter cells are formed.

28 Rabbits and Myxomatosis

- England.
- To hunt for sport.
- Natural predators and selective pressures (such as weather) not found here.
- Rabbits dig underground warrens with many exits, they could come up on the other side of the fence.
- To kill them or drastically reduce the numbers.
- The modern rabbit is resistant to the virus.
- Extremely deadly.
- Yes, the resistant animals reproduced to create a new, resistant, generation or species.
- Teacher to check answers. One suggestion is to introduce a predator (the disadvantage is how the predator will affect other species).

29 Africanis: the world's last pure breed of dog?

- Making something stand out more, improve.
- A secondary effect.
- Originating in a country, where it lives.
- Having a variety of differences.
- Mixed breed dog.
- A group of individuals able to interbreed and produce viable offspring.
- a A specific group or class of animal. b To produce offspring.
- It does not suffer many of the side-effects resulting from inbreeding.
- Teacher to check answers.
- Stereotypes may include: cattle dog (ute owner, labourer), poodle (stuffy, upper-class lady), bulldog (conservative old man).

- Very small dogs to about 3 kg, such as a chihuahua.
- High death rate.
- Undeveloped reproductive systems, tiny size makes it physically stressful.
- Mortality rate of puppies, rarity of breed, degree of difficulty producing litters, 'fashion'.
- Please note that 'positive attributes' are a matter of opinion. Negative attributes are as follows:
 - sinus problems
 - spine problems
 - sinus problems, reduced braincase (can be associated with a drop in intelligence level)
 - stress on skeletal system, heart problems
 - skin rashes, folds can smell where food, etc. gets trapped
 - overprotective even when owner wants others around, dangerous with children.
- Commercial (for profit), contaminated (polluted), brindle (dark streaks, flecks on grey/tawny background), unreliable (cannot be relied on), pure-bred (individuals with near identical genes for a majority of traits), mutation (change in genetic material), aggressive (dangerous, ready to injure), recessive (trait that is only expressed when a dominant alternative is absent).
- Teacher to check. (Re: Approaching breeders, it is not uncommon for breeders to emphasise only the positive traits and avoid discussing negatives.)

30 Evolution and Mutation

- Mitosis and meiosis.
- An alternative form of a gene.
- Causes mutations. Some examples are increased temperature, radiation, chemicals.
- Cancer.
- Sex cell.
- Outward appearance of an individual.
- Fully developed.
- No, less than 1% are beneficial. This is how evolution can occur.
- Teacher to check.
- Those with genes for longer necks will survive to reproduce.
- It is perfectly adapted to its lifestyle.
- Space/the universe.
- Mutation.
- Teacher to check.

31 Genetics Assessment

- A 2 C 3 A 4 C 5 D 6 A 7 A 8 C 9 A 10 C
- D 12 D 13 A 14 C 15 B
- 20
- 38
- 19
- fruit fly
- Small, easy to house and feed, reproduce every 10-14 days so many generations can be studied.
- Alternative forms of a gene.
- Temperature increases, chemicals, radiation.
- For example, colour-blindness, haemophilia.
- Gregor Mendel.
- Meiosis.
- Mitosis.
- True.
- True.
- False.
- True.
- True.

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